STAFFORD'S CONJECTURE ON CYCLICITY OF TORSION MODULES

CHRISTOPHER ALBERT, JOHANNA MOSER AND MAXIMILIAN PHILIPP

ABSTRACT. Let $A_n(k)$ be the Weyl algebra over a field of characteristic zero. We prove that every nonzero $d \in A_n(k)$ admits $F, R, S \in A_n(k)$ such that $1 = dR + FdS$. Thus Stafford's Conjecture 3.8 holds. For $n \geq 1$, ~~The~~ the multiplier F may be chosen as x^N, after a linear symplectic change of variables, where N is the Bernstein degree of d. The proof studies a cyclic quotient through its characteristic variety and excludes nonzero quotients by conormal geometry at infinity.

AI comment Review guide. Red text proposes additions or corrections; blue struck text preserves the original wording. **Johanna:** review the proposals as mathematics and prose, and resolve their boxes after accepting or revising them. No Lean installation, code editing, or proof replay is required for her review. **Max:** review the whole paper–Lean correspondence, including every statement, its hypotheses, and the definitions entering the challenge. The technical supplement contains the comparison notes, mathematical accounts, and optional declaration links. The side-by-side review interface links statements to their checked sources.

1. INTRODUCTION

Let k be a field of characteristic zero, and let

$$A_n(k) = k\langle x_1, \dots, x_n, \partial_1, \dots, \partial_n \rangle$$

be the nth Weyl algebra, with $[\partial_i, x_j] = \delta_{ij}$ and $[\partial_i, \partial_j] = [x_i, x_j] = 0$.
~~Stafford conjectured in 1978 (Conjecture 3.8 of [31, p. 438]) that for every nonzero $d \in A_n(k)$, one can find $F \in A_n(k)$ with $A_n(k) = dA_n(k) + FdA_n(k)$. Stafford notes that this immediately implies $A_n(k)/dA_n(k)$ is cyclic, i.e. generated by a single element. [9] reported the resulting question — whether every right torsion module over $A_n(k)$ is cyclic — as open [Section 3, Conjecture 3.4]. We prove Stafford's conjecture:~~
Stafford's Conjecture 3.8 asks whether every nonzero $d \in A_n(k)$ admits $F \in A_n(k)$ with $A_n(k) = dA_n(k) + FdA_n(k)$ [31, p. 438]. Stafford identifies this with stable cyclicity for finitely generated torsion right Weyl-algebra modules [31, pp. 437–438]. Bellamy's survey states the weaker ordinary-cyclicity question for finitely generated torsion $A_n(\mathbb{C})$-modules as open [9, Section 3, Conjecture 3.4]. We prove Stafford's conjecture:

AI comment INTRO-01. Proposed change — author acceptance pending. The quotient A_n/dA_n is always cyclic, generated by the class of 1. The nontrivial implication concerns general *finitely generated* torsion modules. Bellamy states the ordinary cyclicity question over $\mathbb{C}$; Stafford's stable-cyclicity formulation is stronger. The red replacement above restores these distinctions.

Theorem 1.1. *For every characteristic-zero field k, every $n \geq 1$, and every nonzero $d \in A_n(k)$, there exist $F, R, S \in A_n(k)$ such that*

$$1 = dR + FdS.$$

Moreover, one may choose $F = \ell^{\deg_B d}$ for a linear Weyl coordinate ℓ, obtained by an invertible linear symplectic change of generators. Explicitly, $\ell = \Phi_g^{-1}(x_1)$ for a linear ω-preserving automorphism g as in Section 4.

Formal comparison: Lean also covers $n = 0$, where $A_0(k) = k$

The machine-checked proof of this result is registered in Palomar [7, 8]. Its Global Stafford extension [5, 6] proves the same-divisor identity $1 = dR+FdS$ in $D_k(A)$ for every smooth integral finitely generated characteristic-zero k-algebra A and every nonzero finite-order k-linear differential operator d. Polynomial A recovers the Weyl-algebra identity; the global theorem asserts existence of F, without the fixed linear-coordinate refinement in Theorem 1.1.

Proof sketch: Give each standard generator Bernstein degree one and let $N = \deg_B d$. A linear symplectic change of variables makes d monic of degree N in a momentum p, whose conjugate coordinate we denote by x. It is enough to prove that the cyclic right module

$$Q = A_n/(dA_n + x^N dA_n). \tag{1.1}$$

vanishes. An Euler identity makes x surjective on Q, while monicity makes $H = \{x = 0\}$ noncharacteristic and places $\mathrm{Char}(Q)$ in a hypersurface $V(P)$ with $P(dx) \neq 0$. A filtered-page length argument shows that $\mathrm{Char}(Q)$ has no point over H. Gabber's theorem and the conormal argument of ~~Sections 4 and 5~~ Sections 8 and 9 then force $\mathrm{Char}(Q)$ to be empty. The same argument gives the more precise identity $A_n = dA_n + x^N dA_n$ in the chosen symplectic coordinates.

Constructive predecessors concern Stafford's established generator theorems. Hillebrand–Schmale and Leykin give effective two-generator methods, and Leykin also constructs cyclic generators of holonomic modules [19, 22]. Quadrat–Robertz develop module-decomposition and basis algorithms [27, 28]; Caro–Levcovitz and Caro-Tuesta–Levcovitz study differential-operator rings with series coefficients [11, 12]. The constrained identity for arbitrary nonzero d requires the additional quotient-vanishing argument developed here.

The surrounding generator theory includes Stafford's stable-range and noncommutative Forster–Swan results [33, 32]. The generator article has a corrigendum and related exposition [35, 34]. Stafford also constructed stably free Weyl-algebra ideals [36]. OreModules supplies computational methods for modules over Ore algebras [13]. These provide context; the Euler residue,

filtered-length contradiction, and asymptotic conormal construction below are project arguments.

2. ~~AI Statement [Chris]~~Formal verification

~~Every statement of this paper, including Theorems ?? and 9.3 at the generality stated, is formalized and verified in Lean 4 against Mathlib, with no axiom beyond Lean's three standard ones.~~ The main identity, its fixed-source strengthening and the documented cyclicity and noncharacteristic consequences are verified in Lean 4 against Mathlib using only the three standard axioms. The formal proof also supplies the characteristic-support and geometric results used by its own argument. Statement scope and proof routes must be compared separately: machine verification of a terminal theorem does not certify every intermediate step of the printed proof. The mathematical differences are noted below; full formal comparisons and declaration links are in the technical supplement. The headline theorem is registered in Palomar [7]; the current Lean sources and auxiliary results are archived [2]. The accompanying proof-map supplement names the Lean declaration for each step.

AI comment WF-01. Proposed change — author acceptance pending. Please review each comparison at the stated generality. The Palomar record checks the registered main theorem; later results have separate archived sources. The formal geometric route uses Laurent series, while the printed tangent-limit argument is auxiliary. The short comparison notes identify these differences without claiming that every numbered statement or every proof passage has been verified verbatim.

The development uses Lean 4 [16], Mathlib [37] and AlgebraicAnalysis v0.3.0 [3]; tooling, chronology and statement comparison are described in Appendix A.

Online resources. Online Resource 1 is the interactive paper–Lean correspondence review, including the exact challenge definitions and archived proof endpoints; Online Resource 2 is the readable mathematical account of the selected formal proofs. Both are available from the review companion, with a frozen supplementary package for journal review [1]. The reusable HTML generator is maintained separately under the Apache 2.0 license; its software archive and the manuscript supplement have distinct version records. Proposed corrections and unresolved review actions remain explicitly marked in this draft.

3. Notation and definitions

Throughout, k is a field of characteristic zero and for the n-th Weyl algebra over k, we write

$$A_n(k) = k\langle x_1, \dots, x_n, \partial_1, \dots, \partial_n \rangle, \quad [\partial_i, x_j] = \delta_{ij}, \quad [x_i, x_j] = [\partial_i, \partial_j] = 0.$$

All ideals are *right* ideals and all modules are *right* modules unless the contrary is said explicitly.

Convention 3.1. *We single out the first Weyl pair (x_1, ∂_1) and write*

$$p := \partial_1, \qquad C := A_{n-1}(k)[x_1], \qquad A_n(k) = C[p; \delta],$$

Here $\delta(x_1) = 1$ and δ vanishes on $A_{n-1}(k)$. so that x_1 is central in C, $[p, x_1] = 1$, and the transverse algebra $A_{n-1}(k)$ commutes with x_1 and p. Every element of $A_n(k)$ can uniquely be written as $\sum_j p^j c_j$ with $c_j \in C$.

Definition 3.2 (Filtration [Def. 2.6 and 2.9 [29]])**.** *A filtration $F_\bullet$ on $A_n(k)$ is a sequence of linear subspaces*

$$\{0\} = F_{-1} \subseteq F_0 \subseteq F_1 \subseteq \ldots \subseteq A_n(k),$$

such that $F_i \cdot F_j \subseteq F_{i+j}$ and $A_n(k) = \bigcup F_j$. The associated graded algebra, also called symbol ring, is defined as

$$\operatorname{gr}^F A_n(k) = \bigoplus_{j=0}^{\infty} F_j / F_{j-1}.$$

Let $a \neq 0 \in A_n(k)$ and let j be the smallest integer, such that $a \in F_j$, called a filtration degree $\deg_F a$. *The class of a in F_j/F_{j-1} is called its* symbol $\sigma_j(a)$.

In the proof, we use two filtrations of $A_n(k)$.

- The *Bernstein filtration* $B_\bullet$ assigns degree 1 to every x_i and every ∂_i. The Bernstein degree of d is written $\deg_B d$.
- The *order filtration* $F_\bullet$ assigns degree 0 to every x_i and degree 1 to every ∂_i; F_m is spanned by the monomials $x^\alpha \partial^\beta$ with $|\beta| \leq m$.

Formal comparison: as spans of ordered PBW words of bounded weight

For both the order filtration and the Bernstein filtration, the associated graded algebra $\operatorname{gr} A_n(k)$ is simply the polynomial ring $k[x, \xi] = k[x_1, \ldots, x_n, \xi_1, \ldots, \xi_n]$ in $2n$ variables with ξ_i the symbol of ∂_i. In particular, $\operatorname{gr} A_n(k)$ is commutative [29, Prop. 2.10]. Good filtrations and characteristic varieties are treated systematically for example in [10, 20].

3.1. **Cotangent space and symplectic structure.** Let $X = \mathbb{A}^n_k$ be affine n-space over a field k of characteristic zero, with coordinates $x_1, \ldots, x_n$. Its *cotangent bundle* T^*X is the space of pairs (y, ξ) consisting of a point $y \in X$ together with a covector ξ at y; since X is affine space, this cotangent space is the same vector space $(k^n)^*$ at every point, and we may identify

$$T^*X \;\cong\; X \times (k^n)^*.$$

We write $\xi_1, \ldots, \xi_n$ for the coordinates on $(k^n)^*$ dual to $x_1, \ldots, x_n$, so that a covector is $\sum_i \xi_i \, dx_i$, and identify $\operatorname{gr}^B A_n(k) = k[x, \xi]$ with the coordinate ring of T^*X. We write $\pi \colon T^*X \to X$ for the projection forgetting the covector, and for $y \in X$ write $T^*_y X = \pi^{-1}(y) \cong (k^n)^*$ for the *fibre* of T^*X over y.
We single out the first coordinate x_1 and define the distinguished hyperplane

$$H := \{x_1 = 0\} = \{(0, x_2, \ldots, x_n)\} \subseteq X;$$

its conormal covector is dx_1, i.e. $\xi = (1, 0, \ldots, 0)$.

Definition 3.3. *A filtration of a right A_n-module M compatible with $F_\bullet$ is an exhaustive increasing chain of k-subspaces $M_\bullet$ with $M_iF_j \subseteq M_{i+j}$ and $M_i = 0$ for $i < 0$. It is* good *if the associated graded module $\operatorname{gr} M = \bigoplus_i M_i/M_{i-1}$ is finitely generated over $\operatorname{gr}^F A_n$.*

As an example, the filtration $M_i := qF_i$ of a cyclic module over A_n induced by its generator q is good, since $\operatorname{gr} Q$ is generated by the element $\bar{q}$ denoting the class of q in Q_0/Q_{-1}.
For a finitely generated right A_n-module M with a good order filtration $M_\bullet$, the graded module $\operatorname{gr} M$ is a finitely generated $k[x,\xi]$-module.

Definition 3.4. *The* characteristic variety *(also called symbol variety) of a finitely generated A_n-module M with a good order filtration $M_\bullet$ is defined by*

$$\operatorname{Char}(M) := V\big(\operatorname{Ann}_{k[x,\xi]} \operatorname{gr} M\big) \subseteq T^*\mathbb{A}^n = \operatorname{Spec} k[x,\xi],$$

$\operatorname{Ann}_{k[x,\xi]} \operatorname{gr} M = \{f \in k[x,\xi] : f \cdot m = 0 \,\forall m \in \operatorname{gr} M\}$ *are the* annihilators *of $\operatorname{gr} M$ over $k[x,\xi]$, and V denotes their* vanishing locus*, i.e. the set of points (x,ξ) for which all annihilator polynomials vanish. This set of points lies in the* cotangent bundle *$T^*\mathbb{A}^n$ of the affine n-space $\mathbb{A}^n$.*
$\operatorname{Char}(M)$ is independent of the choice of the good filtration.

Formal comparison: for $M = A_n/I$

Definition 3.5 (Symplectic form and Poisson bracket)**.** *A bilinear form ω on a k-vector space is* symplectic *if it is antisymmetric and nondegenerate. The standard symplectic form on the vector space k^{2n}, with basis $x_1, \dots, x_n$, $\partial_1, \dots, \partial_n$, is determined by the relations*

$$\omega(x_i, \partial_j) = \delta_{ij}, \qquad \omega(x_i, x_j) = \omega(\partial_i, \partial_j) = 0,$$

analogously to the Weyl relations. Identifying $T^*X \cong X \times (k^n)^*$ via the coordinates $x_1, \dots, x_n$ and their duals $\xi_1, \dots, \xi_n$, on T^*X this ω takes the familiar form

$$\omega = \sum_{i=1}^{n} dx_i \wedge d\xi_i.$$

The associated Poisson bracket *on $k[x,\xi]$, the coordinate ring of T^*X, is*

$$\{f, g\} = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i}\frac{\partial g}{\partial \xi_i} - \frac{\partial f}{\partial \xi_i}\frac{\partial g}{\partial x_i} \right).$$

With this convention of the Poisson bracket, the order-filtration symbol of a commutator is $\sigma^F([a,b])$ is $\{\sigma^F(b), \sigma^F(a)\}$.

Definition 3.6 (involutive)**.** *An ideal J is* involutive *if $\{J, J\} \subseteq J$.*

3.2. **Gabber's theorem.**

Theorem 3.7 (Gabber [17][30][20, Theorem 2.3.1 and Appendix D])**.** *Let R be a filtered ring such that $\mathrm{gr}(R)$ is a Noetherian commutative algebra over $\mathbb{Q}$. Let M be a finitely generated R module. Let $J = \sqrt{Ann_{\mathrm{gr}\,R}\,\mathrm{gr}\,M}$ be the radical of the annihilators of M. Then J is involutive, i.e.*

$$\{J, J\} \subset J.$$

Formal comparison: proved in Lean but it is a well known theorem so no need to check the proof

4. Mapping to a monic normal form

Given a fixed nonzero $d \in A_n(k)$ of Bernstein degree N, our first goal is to find a map to replace d by a normal form $\tilde{d}$ that is monic in p, i.e. of the form $\tilde{d} = p^N + \sum_{j<N} p^j a_j$ with coefficients a_j in C.

If $N = 0$, then $d \in B_0 = k$ is a nonzero scalar, hence a unit, and $1 = dR + FdS$ holds trivially with $F = 1 = \ell^0$, $R = d^{-1}$, $S = 0$. We therefore assume $N \geq 1$.

We want to construct a well-behaved automorphism of $\mathrm{gr}^B A_n(k) = k[x, \xi]$ from simpler linear algebra. We proceed in three stages: define an automorphism of the vector space $\mathbb{V} = k^{2n}$, extend it to an automorphism of $A_n(k)$ itself, and observe the induced action this extension leaves on the associated graded ring.

We can identify $\mathbb{V} = k^{2n}$ with the Bernstein-degree-one part B_1/B_0 of $A_n(k)$, i.e. linear combinations of $x_1, \ldots, x_n, \partial_1, \ldots, \partial_n$; the coordinates of a vector are then the coefficients of these generators. By this identification, $\mathbb{V}$ carries the standard symplectic form ω (Def. 3.5).

Lemma 4.1. *Let $(\mathbb{V}, \omega)$ be a symplectic k-space of dimension $2n$ and $0 \neq v \in \mathbb{V}$. Then v is the first vector of a symplectic basis of $\mathbb{V}$; consequently, there exists a k-linear automorphism g preserving ω (i.e. $\omega(gu, gv) = \omega(u, v)$ for all $u, v \in \mathbb{V}$.) with $g(e) = v$ for any prescribed nonzero vector e.*

Proof. Since ω is nondegenerate and $v \neq 0$, the functional $\omega(v, -)$ is not identically zero, so there is $w' \in \mathbb{V}$ with $\omega(v, w') = c \neq 0$; setting $w := w'/c$ gives $\omega(v, w) = 1$.

Let $U = \mathrm{span}(v, w)$. For $0 \neq av + bw \in U$: if $a \neq 0$, $\omega(av + bw, w) = a \neq 0$; if $b \neq 0$, $\omega(av + bw, v) = -b \neq 0$. Either way $av + bw$ pairs nontrivially with something in U, so $\omega|_U$ is nondegenerate.

Since $\omega|_U$ is nondegenerate, $U \cap U^\perp = 0$, and as $\dim U + \dim U^\perp = \dim \mathbb{V}$ for a nondegenerate form, this gives $\mathbb{V} = U \oplus U^\perp$. If $u \in U^\perp$ pairs to zero with all of $U^\perp$, it also pairs to zero with U (as $u \in U^\perp$), hence with all of $\mathbb{V}$, forcing $u = 0$; so $\omega|_{U^\perp}$ is nondegenerate too.

By induction on n (trivial for $n = 1$, where $U = \mathbb{V}$), a symplectic basis e_2, $\ldots, f_n$ of $U^\perp$ exists; setting $e_1 := v$, $f_1 := w$ gives a symplectic basis of

$\mathbb{V} = U \oplus U^\perp$, since cross-pairings between U and $U^\perp$ vanish by definition of $U^\perp$.

Consequently: given nonzero $v, e \in \mathbb{V}$, we can complete both vectors to a symplectic basis as above. We then define the linear map g to send e to v, and each subsequent e-basis vector to the corresponding v-basis vector, thus preserving all pairwise values of ω on basis vectors. Through the bilinearity of ω, g then preserves ω on all of $\mathbb{V}$; and $g(e) = v$. □

Let g be a k-linear automorphism of $\mathbb{V}$ that preserves ω. Since $\mathbb{V}$ is identified with the span of the generators of $A_n(k)$, g assigns to each generator u a linear combination $g(u)$ of generators. Because $A_n(k)$ is presented by generators and the Weyl relations $[v, u] = \omega(u, v)$, and g preserves ω, the elements $g(u)$ satisfy the same relations as the u. Hence the substitution $u \mapsto g(u)$ extends to a unique k-algebra endomorphism Φ_g of $A_n(k)$. Because g is an automorphism, so is Φ_g, and we have $\Phi_g^{-1} = \Phi_{g^{-1}}$.

Proposition 4.2. *Let $0 \neq d \in A_n(k)$ with $N := \deg_B d \geq 1$. There exists an automorphism Φ_g of $A_n(k)$ and $c \in k^\times$ such that*

$$\tilde{d} := c^{-1}\Phi_g(d) = p^N + \sum_{j<N} p^j a_j, \qquad a_j \in C, \quad \deg_B a_j \leq N - j \tag{4.1}$$

in the notation of Convention 3.1. $\tilde{d}$ itself has $\deg_B \tilde{d} = N$.

Proof. Let $s \in k[x, \xi]$ be the Bernstein principal symbol of d, a nonzero homogeneous polynomial of degree N on $\mathbb{V}$. A characteristic-zero field is infinite, so every nonzero polynomial has a nonvanishing value: we choose $v \in \mathbb{V}$ with $s(v) \neq 0$. Because s is homogeneous of degree $N \geq 1$, $s(0)$ must vanish, therefore v is nonzero. Let $e_p \in \mathbb{V}$ be the standard basis vector corresponding to p. By Lemma 4.1 there is an ω-preserving automorphism h of $\mathbb{V}$ with $h(e_p) = v$. The standard basis is symplectic, so the Gram matrix J of ω in it satisfies $J^{-1} = -J$; hence $h^\mathsf{T} J h = J$ implies $hJh^\mathsf{T} = J$, i.e. the transpose $g := h^\mathsf{T}$ is again ω-preserving. Let Φ_g be the corresponding automorphism of $A_n(k)$, extending $u \mapsto g(u)$ on generators. Let M be the matrix whose rows are the images of the generators, i.e. $g(u_i) = \sum_j M_{ij} u_j$. Since Φ_g sends each generator to a degree-one element, it maps B_m into B_m for all m, and since Bernstein symbols are multiplicative, a standard monomial of degree N is mapped to an element of B_N whose class modulo B_{N-1} is obtained by substituting $u_i \mapsto \sum_j M_{ij} u_j$ in the corresponding commutative monomial. Hence $\Phi_g(d) \equiv s(Mu) \bmod B_{N-1}$. As M is invertible, $s(Mu)$ is again a nonzero homogeneous polynomial of degree N, so $\deg_B \Phi_g(d) = N$ and the Bernstein symbol of $\Phi_g(d)$ is the function $s_g \colon a \mapsto s(Ma)$. Because $g = h^\mathsf{T}$, the rows of M are the columns of h, i.e. $Me_j = h(e_j)$. Hence the value of s_g at e_p is

$$s_g(e_p) = s(Me_p) = s(h(e_p)) = s(v) \neq 0.$$

Set $c := s(v)$. Since c is a nonzero scalar, it is a central unit of $A_n(k)$, and $\tilde{d} := c^{-1}\Phi_g(d)$ satisfies $\deg_B \tilde{d} = N$ and $\sigma^B(\tilde{d})(e_p) = 1$. Using the normal

form $A_n(k) = \bigoplus_j p^j C$, we can write $\tilde{d} = \sum_{j=0}^{N} p^j a_j$ with $a_j \in C$. The Bernstein filtration is compatible with this decomposition, so $N = \deg_B \tilde{d} = \max_j (j + \deg_B a_j)$, and we have $\deg_B a_j \leq N - j$ for all j; in particular $a_j = 0$ for $j > N$ and $a_N \in C \cap B_0 = k$. The symbol is

$$\sigma^B(\tilde{d}) = \sum_{j+\deg_B a_j = N} \sigma^B(p)^j \, \sigma^B(a_j),$$

For $j < N$ the factors $\sigma^B(a_j)$ must have a positive degree and must therefore be non-constant in a coordinate other than p, so they vanish at e_p. Because we defined $\tilde{d}$ to be $c^{-1}\Phi_g(d)$, evaluating at e_p gives $a_N = \sigma^B(\tilde{d})(e_p) = c^{-1}\sigma^B(\Phi_g(d))(e_p) = c^{-1}s_g(e_p) = c^{-1}s(v) = 1$, and therefore

$$\tilde{d} = p^N + \sum_{j<N} p^j a_j, \qquad a_j \in C, \quad \deg_B a_j \leq N - j.$$

□

Formal comparison: Actual form is not proved in lean, what is important and subsequently used is the fact that the symbol is monic in p and its coefficient is 1. ("PBWmonicity")

Remark 4.3. Since Φ_g is a ring automorphism, so an identity $1 = \tilde{d}R + x_1^N \tilde{d}S$ would transport back to

$$1 = d\big(c^{-1}\Phi_g^{-1}(R)\big) + F\, d\big(c^{-1}\Phi_g^{-1}(S)\big), \qquad F = \Phi_g^{-1}(x_1)^N.$$

Formal comparison: compare

We now use the order filtration on $A_n(k)$ (where coordinates have degree zero and derivatives have degree one) to characterize $P = \sigma_{\mathrm{ord}}(\tilde{d})$, the order principal symbol of $\tilde{d}$. Recall that a symbol, being the class of an element in a single graded piece, is automatically *homogeneous* of the corresponding degree, i.e. contains only terms of that total degree.

Definition 4.4. *We say a polynomial $P = \sum_{i=0}^{N} c_i \xi_1^i \in k[\xi_1, \ldots, \xi_n]$, written as a polynomial in ξ_1 with coefficients $c_i \in k[\xi_2, \ldots, \xi_n]$, is* monic of degree N in ξ_1 *if $c_N = 1$.*

Lemma 4.5. *Let $\tilde{d} = p^N + \sum_{j<N} p^j a_j$ with $a_j \in C$, $\deg_B a_j \leq N - j$, as in (4.1), and set $N = \deg_B \tilde{d}$. Then $\tilde{d}$ has order N, and its order principal symbol is a polynomial in ξ alone,*

$$\sigma_{\mathrm{ord}}(\tilde{d}) = P(\xi), \qquad P \in k[\xi_1, \ldots, \xi_n], \qquad \deg P = N. \tag{4.2}$$

P is monic of degree N in ξ_1 and in particular

$$P(1, 0, \ldots, 0) = 1. \tag{4.3}$$

Proof. Since $\deg_B \tilde{d} = N$, every PBW monomial $x^\alpha \partial^\beta$ occurring in $\tilde{d}$ satisfies $|\alpha| + |\beta| \leq N$. A monomial contributing to the order-N part has $|\beta| = N$, hence $|\alpha| = 0$: it is a pure ∂^β with scalar coefficient. Thus $\tilde{d}$ has order N, and its order-N part is a k-linear combination of ∂^β with $|\beta| = N$, giving $\sigma_{\mathrm{ord}}(\tilde{d}) = P(\xi)$, homogeneous of degree N with coefficients in k.
The monomial $p^N = \partial_1^N$ occurs in $\tilde{d}$ with coefficient 1; every other order-N monomial comes from some $p^j a_j$ with $j < N$ and $a_j \in C$ (hence involving no further factor of $p = \partial_1$), so it carries a positive power of some transverse ∂_i, $i > 1$. Such a monomial contributes only to coefficients c_i with $i < N$, never to c_N; hence c_N receives no contribution besides the coefficient of p^N itself, giving $c_N = 1$ identically, i.e. P is monic of degree N in ξ_1. Evaluating at $(1, 0, \ldots, 0)$ then gives $P(1, 0, \ldots, 0) = 1$. □

5. x_1 AS A SURJECTIVE MAP

We define $E = x_1 p$ and

$$I = \tilde{d} A_n + x_1^N \tilde{d} A_n, \qquad Q = A_n / I. \tag{5.1}$$

The goal of this section will be to show that $Q = Q x_1$ (Proposition 5.3).
The commutator of E defines a grading of A_n: x_1 has Euler degree 1, p has Euler degree -1, and the transverse Weyl algebra has Euler degree zero. Let $A_{\geq 0}$ denote the nonnegative part of A_n in this grading. In PBW form (with x_i on the left of ∂_i), $A_{\geq 0}$ is generated by the transverse algebra, E, and x_1. By the commutation relations of the Weyl algebra, it holds that

$$\begin{aligned} E x_1 &= x_1 (E + 1), \qquad & x_1 E &= (E - 1) x_1, \\ E p &= p (E - 1), \qquad & p E &= (E + 1) p \end{aligned} \tag{5.2}$$

Additionally, x_1 commutes with the transverse algebra, so x_1 is normal in $A_{\geq 0}$. This also means that every element of Euler degree ≥ 1 lies in $A_{\geq 0} x_1$.

Formal comparison: Lean uses the subring generated by the transverse algebra, x_1 and $x_1 p$, and one-sided normality $x_1 r = r' x_1$, which is all that is used

Lemma 5.1. *For every $m \geq 0$,*

$$x_1^m p^m = \prod_{i=0}^{m-1} (E - i), \qquad p^m x_1^m = \prod_{i=1}^{m} (E + i), \tag{5.3}$$

the factors in each product commuting with one another.

Proof. Since both products in (5.3) are polynomials in the single element E, their factors commute. We induct on m; both identities are trivial for $m = 0$.

Assuming the first identity for m,

$$x_1^{m+1}p^{m+1} = x_1\Big(\prod_{i=0}^{m-1}(E-i)\Big)p = \Big(\prod_{i=0}^{m-1}(E-1-i)\Big)x_1 p$$
$$= \Big(\prod_{i=1}^{m}(E-i)\Big)E = \prod_{i=0}^{m}(E-i),$$

using $x_1 E = (E-1)x_1$ repeatedly and the definition of E. The second identity follows identically from $Ep = p(E-1)$. □

Lemma 5.2. *There exists an element $U \in A_{\geq 0}$ such that*

$$1 + Ux_1 \in I. \tag{5.4}$$

Proof. From Lemma 5.1 we define

$$p^N x_1^N = \prod_{r=1}^{N}(E+r) =: R_+(E), \qquad x_1^N p^N = \prod_{r=0}^{N-1}(E-r) =: R_-(E).$$

The roots of R_+ and R_- are disjoint (the sets $\{-1,\dots,-N\}$ and $\{0,\dots,N-1\}$ of integers, embedded in the prime field $\mathbb{Q} \subset k$), so $\gcd(R_+, R_-) = 1$ in $\mathbb{Q}[E]$. By the extended Euclidean algorithm, there exist $u, v \in \mathbb{Q}[E] \subset k[E]$ with

$$R_+(E)u(E) + R_-(E)v(E) = p^N x_1^N u(E) + x_1^N p^N v(E) = 1.$$

We write $\tilde{d} = p^N + \Sigma$ with $\Sigma = \sum_{j<N} p^j a_j$. Then

$$(\tilde{d}\, x_1^N)u(E) + (x_1^N \tilde{d})v(E) = 1 + w \in I,$$

where

$$w = \Sigma x_1^N u(E) + x_1^N \Sigma v(E).$$

We can write each a_j in Σ in powers of x_1, with coefficients in the transverse Weyl algebra. A term $p^j x_1^r b$ coming from $p^j a_j$ has $r \geq 0$ and $j < N$, and multiplication by x_1^N gives that term Euler degree at least $N - j > 0$. u and v only depend on E, so they have Euler degree 0. Therefore every term of w has positive Euler degree, and thus $w \in A_{\geq 0}x_1$ - or equivalently, there exists a $U \in A_{\geq 0}$, such that $w = Ux_1$. □

Formal comparison: the ring identity is Lean's; the factorization uses left coefficients $c\,x^a p^j$ but that does not change anything because Euler degree does not depend on the order of x and p

Proposition 5.3. *Right multiplication by x_1 on Q is surjective.*

Proof. Let q be the class of 1 in Q. By Equation (5.4), we have $q = -qUx_1$, so $q \in (qA_{\geq 0})x_1$. We use the normality of x_1 to show

$$qA_{\geq 0} = (qA_{\geq 0})x_1. \tag{5.5}$$

The inclusion $\supseteq$ holds because $x_1 \in A^{\geq 0}$ and $A^{\geq 0}$ is a ring. For $\subseteq$, let $r \in A^{\geq 0}$ and choose r' with $x_1 r = r' x_1$ (possible since x_1 is normal) then $qr = -(qU)x_1 r = -(qUr')x_1 \in (qA^{\geq 0})x_1$ since $Ur' \in A^{\geq 0}$.
$qA_{\geq 0}$ is stable under right multiplication by p: If $m = m'x_1$, then

$$mp = m'(x_1 p) = m'E \in qA_{\geq 0}.$$

It is of course also stable under C, since $C = A_{n-1}(k)[x_1] \subseteq A_{\geq 0}$ because no element of C contains p. Since C and p together generate A_n, we have $qA_{\geq 0} = qA_n = Q$. Equation (5.5) is therefore $Q = Qx_1$. □

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6. Vanishing cokernel

Definition 6.1 ([38][Tag 00CM]). *Let R be a commutative ring and $S \subseteq R$ a multiplicative subset (i.e. $1 \in S$ and S is closed under products). The* localization $S^{-1}R$ *is the ring of fractions a/s ($a \in R$, $s \in S$), where*

$$\frac{a}{s} = \frac{a'}{s'} \iff t(s'a - sa') = 0 \quad \text{for some } t \in S,$$

with the usual addition and multiplication of fractions. For an R-module N, the localization $S^{-1}N$ *is the $S^{-1}R$-module of fractions y/s ($y \in N$, $s \in S$), with $y/s = y'/s'$ iff $t(s'y - sy') = 0$ for some $t \in S$. For a prime $\mathfrak{p} \subset R$ we write $R_\mathfrak{p} = S^{-1}R$ and $N_\mathfrak{p} = S^{-1}N$ for $S = R \setminus \mathfrak{p}$. Localization is exact: it sends exact sequences of R-modules to exact sequences of $S^{-1}R$-modules.*

Definition 6.2 ([38][Definition 10.40.1]). *Let R be a ring and let M be an R-module. The* support *of M is the set*

$$\operatorname{Supp}_R(M) = \{\mathfrak{p} \in \operatorname{Spec}(R) | M_\mathfrak{p} \neq 0\}.$$

For this section, we write $M = \operatorname{gr} Q$ and denote the right multiplication by x_1 by $f : M \to M$. We furthermore define

$$V = \ker(f), \qquad U = \operatorname{coker}(f) = M/x_1 M$$
$$T = k[x_2, \ldots, x_n, \xi_2, \ldots, \xi_n],$$

hence $T[x_1, \xi_1] = k[x, \xi]$. If $n = 1$, we define $T = k$. We want to prove that $U = 0$ by comparing two lengths over a localization of T.

Definition 6.3 ([15, 14], [38, Tags 02LY, 00J3, 00IV]). *The length ℓ_T of a T-module of finite length N is defined as the number of steps in a composition series of N such that*

$$0 = N_0 \subsetneq N_1 \subsetneq \cdots \subsetneq N_{\ell_T} = N$$

where each successive quotient N_i/N_{i-1} is simple (i.e. has no proper nontrivial submodules). The length is well-defined, as it does not depend on the choice of the composition series. Lengths are additive for short exact sequences.

Lemma 6.4. *Let R be a commutative ring, N an R-module and $\mathfrak{p} \subset R$ a prime. Then $\mathfrak{p} \in \operatorname{Supp}_R(N)$ if and only if there is $y \in N$ with $\operatorname{Ann}_R(y) \subseteq \mathfrak{p}$.*

[]

Proof. In $N_{\mathfrak{p}}$, a fraction is zero when it can be cleared by an element outside $\mathfrak{p}$. For $y/1$ this means: there is $u \in R \setminus \mathfrak{p}$ with $uy = 0$. Such a u lies in $\operatorname{Ann}_R(y)$ and not in $\mathfrak{p}$. So

$$y/1 = 0 \iff \operatorname{Ann}_R(y) \not\subseteq \mathfrak{p},$$

and, taking the contrapositive,

$$y/1 \neq 0 \iff \operatorname{Ann}_R(y) \subseteq \mathfrak{p}.$$

($\Leftarrow$) Suppose $y \in N$ has $\operatorname{Ann}_R(y) \subseteq \mathfrak{p}$. Then, $y/1 \neq 0$ in $N_{\mathfrak{p}}$. So $N_{\mathfrak{p}}$ has a nonzero element, hence $N_{\mathfrak{p}} \neq 0$ and $\mathfrak{p} \in \operatorname{Supp}_R(N)$.
($\Rightarrow$) Suppose $N_{\mathfrak{p}} \neq 0$. Pick a nonzero element of $N_{\mathfrak{p}}$. It has the form y/s with $y \in N$ and $s \notin \mathfrak{p}$. The element s is invertible in $N_{\mathfrak{p}}$, so

$$y/s = 0 \iff y/1 = 0.$$

Since $y/s \neq 0$, we get $y/1 \neq 0$, and therefore $\operatorname{Ann}_R(y) \subseteq \mathfrak{p}$, which is the desired y. □

Lemma 6.5. *The modules U and V are finite over T, and $\operatorname{Supp}_T V \subseteq \operatorname{Supp}_T U$. If $\mathfrak{q}$ is a minimal prime in $\operatorname{Supp}_T U$, then $U_{\mathfrak{q}}$ and $V_{\mathfrak{q}}$ have finite length over $T_{\mathfrak{q}}$.*

Proof. The good quotient filtration makes $M = \operatorname{gr} Q$ finitely generated over $\operatorname{gr} A_n = k[x, \xi] = T[x_1, \xi_1]$. Since P annihilates M (Corollary 7.1), this action factors through $T[x_1, \xi_1]/(P)$, which, because P is monic of degree N in ξ_1, is generated over $T[x_1]$ by $1, \xi_1, \ldots, \xi_1^{N-1}$. Thus M is finite over $T[x_1]$. Because M is finite, and $T[x_1]$ is a Noetherian ring, the kernel V and cokernel U of multiplication in M by x_1 are also finite over $T[x_1]$. Since they are annihilated by x_1, they are finite over T.
Let $\mathfrak{p} \in \operatorname{Supp}_T(V)$. By Lemma 6.4, we can choose $v \in V = \ker f$ with $\operatorname{Ann}_T(v) \subseteq \mathfrak{p}$. Let us now consider the annihilators of v in $k[x, \xi] = T[x_1, \xi_1]$, defining S as the image of $T \setminus \mathfrak{p}$ in $k[x, \xi]$: $\operatorname{Ann}_{k[x,\xi]}(v) \cap S = \varnothing$, because were $r \in T \setminus \mathfrak{p}$ and in $\operatorname{Ann}_{k[x,\xi]}(v)$, then v would be annihilated by r, which in turn would imply that r is also in $\operatorname{Ann}_T(v)$. Since $\operatorname{Ann}_T(v) \subseteq \mathfrak{p}$, but we assumed $r \in T \setminus \mathfrak{p}$, this is a contradiction. There must therefore exist a prime $\mathfrak{q} \in k[x, \xi]$ with $\mathfrak{q} \cap S = \varnothing$ and $\operatorname{Ann}_{k[x,\xi]}(v) \subseteq \mathfrak{q}$. Since $v \in V = \ker f$, by Lemma 6.4, it follows that $\mathfrak{q} \in \operatorname{Supp}_{k[x,\xi]}(\ker f)$. From Vasconcelos' theorem [23, Theorem 2.4] follows that $\operatorname{Supp}_{k[x,\xi]}(\ker f) \subseteq \operatorname{Supp}_{k[x,\xi]}(\operatorname{coker} f)$: We prove the contrapositive. Let $\mathfrak{q}$ be a prime of $k[x, \xi]$ with $\mathfrak{q} \notin \operatorname{Supp}_{k[x,\xi]}(\operatorname{coker} f)$, i.e. $(\operatorname{coker} f)_{\mathfrak{q}} = 0$. Since localization is exact, $(\operatorname{coker} f)_{\mathfrak{q}} = \operatorname{coker}(f_{\mathfrak{q}})$, so $\operatorname{coker}(f_{\mathfrak{q}}) = 0$: the endomorphism $f_{\mathfrak{q}} \colon M_{\mathfrak{q}} \to M_{\mathfrak{q}}$ is therefore surjective. The localization $M_{\mathfrak{q}}$ is a finite module over the Noetherian ring $k[x, \xi]_{\mathfrak{q}}$, and $f_{\mathfrak{q}}$ is $k[x, \xi]_{\mathfrak{q}}$-linear, because f is multiplication by $x_1 \in k[x, \xi]$ and $k[x, \xi]$ is commutative. By Vasconcelos' theorem, a surjective endomorphism of a finite module is injective. Hence $f_{\mathfrak{q}}$ is injective, and by exactness of localization, $(\ker f)_{\mathfrak{q}} = \ker(f_{\mathfrak{q}}) = 0$, i.e. $\mathfrak{q} \notin \operatorname{Supp}_{k[x,\xi]}(\ker f)$. The contrapositive is

therefore $\mathfrak{q} \in \mathrm{Supp}_{k[x,\xi]}(\ker f)$, then $\mathfrak{q} \in \mathrm{Supp}_{k[x,\xi]}(\mathrm{coker}\, f) = \mathrm{Supp}_{k[x,\xi]}(U)$. By Lemma 6.4 there therefore exists an $u \in U$ with $\mathrm{Ann}_{k[x,\xi]}(u) \subseteq \mathfrak{q}$. To get back to the support over T, we define $r \in \mathrm{Ann}_T(u)$. Then $ru = 0$, so $r \in \mathrm{Ann}_{k[x,\xi]}(u) \subseteq \mathfrak{q}$. If $r \notin \mathfrak{p}$, the image of r would lie in $\mathfrak{q} \cap S = \varnothing$, which is impossible. Hence $r \in \mathfrak{p}$, i.e. $\mathrm{Ann}_T(u) \subseteq \mathfrak{p}$, and Lemma 6.4 gives $\mathfrak{p} \in \mathrm{Supp}_T(U)$. Since $\mathfrak{p} \in \mathrm{Supp}_T(V)$ was arbitrary, $\mathrm{Supp}_T(V) \subseteq \mathrm{Supp}_T(U)$. □

AI comment ALG-04. Max — correspondence check; no rewrite of this lemma requested. Check the monic quotient argument giving finite T-modules U, V, together with the subsequent T-action and T-linear page maps. Localization applies to the associated graded pages. See COMP-06.

Lemma 6.6. *For every minimal prime* $\mathfrak{q}$ *in* $\mathrm{Supp}_T U$,

$$\ell_{T_\mathfrak{q}}(U_\mathfrak{q}) \leq \ell_{T_\mathfrak{q}}(V_\mathfrak{q}).$$

Proof. To prove the inequality we set up a spectral sequence measuring how $\ker(f)$ and $\mathrm{coker}(f)$ acting on Q itself compare, order by order, to V and U, acting on $\mathrm{gr}\, Q$.

Consider the filtered complex $Q \xrightarrow{f} Q$, with decreasing filtration $G^p = F_{-p}Q$, where F is the order filtration. Since x_1 has order 0, $f(G^p) \subseteq G^p$ for every p; the induced map on G^p/G^{p+1}, summed over p, is exactly the action of x_1 on $M = \mathrm{gr}\, Q$. This is the crudest approximation to $\ker_Q(f)$ and $\mathrm{coker}_Q(f)$: it correctly identifies V, U as approximations, but it does not yet see elements that vanish to higher order than this naive count predicts. The spectral sequence refines this approximation in successive stages, called *pages*: for each $r \geq 0$, the source and target pages are

$$A_r = \bigoplus_{p\in\mathbb{Z}} \frac{G^p \cap f^{-1}(G^{p+r})}{G^{p+1} \cap f^{-1}(G^{p+r})}, \qquad B_r = \bigoplus_{p\in\mathbb{Z}} \frac{G^p}{(G^p \cap f(G^{p-r+1})) + G^{p+1}},$$

where the degree-p summand of A_r isolates elements of G^p whose image under f drops r extra filtration levels (candidates for the kernel, accurate to order r), and the degree-p summand of B_r isolates the part of G^p not yet hit by f from $r-1$ levels shallower.

Applying f to a representative gives a well-defined map $d_r : A_r \to B_r$, shifting the summand index by r; one checks directly from these definitions that

$$A_0 = B_0 = M, \qquad d_0 = x_1, \qquad A_{r+1} = \ker d_r, \qquad B_{r+1} = \mathrm{coker}\, d_r,$$

so that each page is obtained from the previous one by taking kernel and cokernel of d_r, exactly as in the construction of a spectral sequence from a filtered complex.

These pages carry a T-module structure. The pages A_r, B_r are subquotients of Q, which a priori carries an action of the transverse Weyl algebra $A_{n-1}(k)$ by right multiplication. This action commutes with f, since the transverse algebra commutes with x_1, so the maps d_r are compatible with it. Two transverse elements commute only up to a commutator, which lowers the

order filtration by one. On the page quotients such lower-order terms vanish, so the induced actions commute, and the commutative ring T acts on every A_r and B_r, via the symbols of the transverse generators. The maps d_r and the successor-page maps are T-linear. In particular, this means that localizing at $T \setminus \mathfrak{q}$ and measuring lengths over $T_{\mathfrak{q}}$ is a well-defined action for the pages. Now we fix a minimal prime $\mathfrak{q}$ in $\operatorname{Supp}_T U$ and localize at $T\setminus\mathfrak{q}$. We localize the T-modules A_r, B_r and their T-linear maps. Exactness of localization preserves the successor-page kernel and cokernel identities. By Lemma 6.5, $(A_1)_{\mathfrak{q}} = V_{\mathfrak{q}}$ and $(B_1)_{\mathfrak{q}} = U_{\mathfrak{q}}$ have finite length over $T_{\mathfrak{q}}$; since each $(A_{r+1})_{\mathfrak{q}} \subseteq (A_r)_{\mathfrak{q}}$ and $(B_{r+1})_{\mathfrak{q}}$ is a quotient of $(B_r)_{\mathfrak{q}}$, all later pages have finite length as well.
The maps $(B_1)_{\mathfrak{q}} \to (B_r)_{\mathfrak{q}}$ are surjective by construction. Let $K_r := \ker((B_1)_{\mathfrak{q}} \to (B_r)_{\mathfrak{q}})$. Since $(B_1)_{\mathfrak{q}} \to (B_{r+1})_{\mathfrak{q}}$ factors through $(B_1)_{\mathfrak{q}} \to (B_r)_{\mathfrak{q}} \to (B_{r+1})_{\mathfrak{q}}$, we have $K_r \subseteq K_{r+1}$, so $(K_r)_{r\geq 1}$ is an increasing chain of submodules of $(B_1)_{\mathfrak{q}}$. This chain exhausts $(B_1)_{\mathfrak{q}}$: Before localization, take a homogeneous class in B_1, represented by $z \in G^p$. Since $Q = Qx_1$, choose one preimage $z = wx_1$. Exhaustiveness of the filtration means that w must be in G^s for some s; choose $r \geq 1$ with $p - r + 1 \leq s$. Then $w \in G^{p-r+1}$, so z belongs to the denominator of B_r and its class vanishes. Every class is a finite sum of homogeneous classes and hence vanishes on a sufficiently late page. The same holds for a localized class z/a, since its numerator already vanishes on such a page. As $z \in B_1$ was arbitrary, so is z/a in $(B_1)_{\mathfrak{q}}$, and $\bigcup_{r\geq 1} K_r = (B_1)_{\mathfrak{q}}$. Since $(B_1)_{\mathfrak{q}}$ has finite length, it is a Noetherian $T_{\mathfrak{q}}$-module. By the ascending chain condition, the increasing chain $K_1 \subseteq K_2 \subseteq \cdots$ stabilizes: there exists a finite R with $K_R = K_{R+i}$ for all $i \geq 0$. Combined with $\bigcup_r K_r = (B_1)_{\mathfrak{q}}$, stabilization forces $K_R = (B_1)_{\mathfrak{q}}$. Finally, $K_R = (B_1)_{\mathfrak{q}}$ means $(B_1)_{\mathfrak{q}} \to (B_R)_{\mathfrak{q}}$ must be the zero map: by the first isomorphism theorem, $(B_1)_{\mathfrak{q}}/K_R \cong \operatorname{im}((B_1)_{\mathfrak{q}} \to (B_R)_{\mathfrak{q}})$, and $K_R = (B_1)_{\mathfrak{q}}$ gives $(B_1)_{\mathfrak{q}}/K_R = 0$, so $\operatorname{im}((B_1)_{\mathfrak{q}} \to (B_R)_{\mathfrak{q}}) = 0$. But $(B_1)_{\mathfrak{q}} \to (B_R)_{\mathfrak{q}}$ is also surjective, so $(B_R)_{\mathfrak{q}} = 0$.

At each stage $r \geq 1$ the differential $(d_r)_{\mathfrak{q}}$ gives an exact sequence

$$0 \to (A_{r+1})_{\mathfrak{q}} \to (A_r)_{\mathfrak{q}} \xrightarrow{(d_r)_{\mathfrak{q}}} (B_r)_{\mathfrak{q}} \to (B_{r+1})_{\mathfrak{q}} \to 0,$$

which splits at $\operatorname{im}((d_r)_{\mathfrak{q}})$ into two short exact sequences. Additivity of length gives $\ell((A_r)_{\mathfrak{q}}) = \ell((A_{r+1})_{\mathfrak{q}}) + \ell(\operatorname{im}(d_r)_{\mathfrak{q}})$ and $\ell((B_r)_{\mathfrak{q}}) = \ell(\operatorname{im}(d_r)_{\mathfrak{q}}) + \ell((B_{r+1})_{\mathfrak{q}})$; eliminating $\ell(\operatorname{im}(d_r)_{\mathfrak{q}})$ yields, for every $r \geq 1$,

$$\ell((B_r)_{\mathfrak{q}}) + \ell((A_{r+1})_{\mathfrak{q}}) = \ell((A_r)_{\mathfrak{q}}) + \ell((B_{r+1})_{\mathfrak{q}}). \tag{6.1}$$

Summing this identity over $r = 1, \ldots, R-1$ and reindexing the sums, most terms cancel on both sides, leaving

$$\ell((B_1)_{\mathfrak{q}}) + \ell((A_R)_{\mathfrak{q}}) = \ell((A_1)_{\mathfrak{q}}) + \ell((B_R)_{\mathfrak{q}}).$$

Since $(B_R)_{\mathfrak{q}} = 0$ and $(A_1)_{\mathfrak{q}} = V_{\mathfrak{q}}$, $(B_1)_{\mathfrak{q}} = U_{\mathfrak{q}}$, this reads

$$\ell(U_{\mathfrak{q}}) + \ell((A_R)_{\mathfrak{q}}) = \ell(V_{\mathfrak{q}}).$$

As $\ell((A_R)_\mathfrak{q}) \geq 0$, we conclude

$$\ell_{T_\mathfrak{q}}(U_\mathfrak{q}) \leq \ell_{T_\mathfrak{q}}(V_\mathfrak{q}).$$

□

[Readable correspondence note.]
On the other hand, we can also establish a different inequality.

Lemma 6.7. *For every minimal prime* $\mathfrak{q}$ *in* $\operatorname{Supp}_T U$*,*

$$\ell_{T_\mathfrak{q}}(U_\mathfrak{q}) > \ell_{T_\mathfrak{q}}(V_\mathfrak{q}).$$

Proof. Let us consider $\operatorname{Ann}_C M$. By Gabber's Theorem (3.7, [17]), $I = \sqrt{\operatorname{Ann}_C M}$ is involutive. If $\mathfrak{p}$ is a minimal prime of J, then $\mathfrak{p}$ is itself involutive, i.e. $\{\mathfrak{p}, \mathfrak{p}\} \subseteq \mathfrak{p}$. We can show this the following way: We localize $k[x, \xi]$ at $\mathfrak{p}$. Since $\mathfrak{p}$ is minimal over I, the only prime of $k[x, \xi]_\mathfrak{p}$ containing $I_\mathfrak{p}$ is $\mathfrak{p}_\mathfrak{p}$ itself; as I is already radical, so is its localization, giving

$$I_\mathfrak{p} = \sqrt{I_\mathfrak{p}} = \mathfrak{p}_\mathfrak{p}.$$

The Poisson bracket extends to $k[x, \xi]_\mathfrak{p}$ since by the Leibniz rule, $\{a/s,\, b/t\}$ is determined by $\{a, b\}$ and the brackets of a, b with s, t, giving a well-defined bracket on $k[x, \xi]_\mathfrak{p}$ agreeing with the one on $k[x, \xi]$ under $f \mapsto f/1$; in particular $\{f/1, g/1\} = \{f, g\}/1$ for $f, g \in k[x, \xi]$, and the involutivity of I localizes to $\{I_\mathfrak{p}, I_\mathfrak{p}\} \subseteq I_\mathfrak{p}$. Now take $f, g \in \mathfrak{p}$. Then $f/1, g/1 \in \mathfrak{p}_\mathfrak{p} = I_\mathfrak{p}$, so

$$\{f, g\}/1 = \{f/1, g/1\} \in I_\mathfrak{p} = \mathfrak{p}_\mathfrak{p}.$$

Thus there is $t \notin \mathfrak{p}$ with $t\{f, g\} \in \mathfrak{p}$. Since $\mathfrak{p}$ is prime and $t \notin \mathfrak{p}$, this forces $\{f, g\} \in \mathfrak{p}$.
From the involutivity of $\mathfrak{p}$ follows that no $\mathfrak{p}$ contains x_1: Assume $x_1 \in \mathfrak{p}$. Since $P \in \mathfrak{p}$, also the Poisson brackets

$$\{x_1, P\}, \quad \{x_1, \{x_1, P\}\}, \ldots$$

lie in $\mathfrak{p}$. $\{x_1, -\}$ differentiates once with respect to ξ_1, and because P is homogeneous of degree N and monic in ξ_1, its Nth derivative is the nonzero constant $N!$ - therefore the Nth bracket would put $N!$ in $\mathfrak{p}$. Since $\mathfrak{p}$ is a proper prime ideal, no nonzero constant can be in $\mathfrak{p}$, so we must conclude $x_1 \notin \mathfrak{p}$. This result holds under localization at $T \setminus \mathfrak{q}$: no minimal prime of $\operatorname{Ann}_{k[x,\xi]_\mathfrak{p}}(M_\mathfrak{q})$ contains x_1 either, since the annihilator of the finite module M commutes with localization. Consider the increasing chain of kernels of powers of the localization $f_\mathfrak{q}$ of right multiplication with x_1 $\ker(f_\mathfrak{q}) \subseteq \ker(f_\mathfrak{q}^2) \subseteq \ldots$. Since $M_\mathfrak{q}$ is Noetherian, this chain stabilizes. Let $K := \ker(f_\mathfrak{q}^m)$ at a stage where it stabilizes, so $f_\mathfrak{q}(K) = x_1 K \subseteq K$ and $\ker(f_\mathfrak{q}^{m+1}) = \ker(f_\mathfrak{q}^m) = K$. Each quotient $\ker(f_\mathfrak{q}^{i+1})/\ker(f_\mathfrak{q}^i)$ maps to $V_\mathfrak{q} = \ker(f_\mathfrak{q})$ injectively via $v \mapsto {x_1}^i v$, so, as $V_\mathfrak{q}$ has finite length, so does each quotient and hence K itself.
Two elementary facts about K:

- ${f_\mathfrak{q}}^{-1} K = K$: if $v \in K$ then $f_\mathfrak{q}(v) \in K$ (as $f_\mathfrak{q}(K) \subseteq K$); if $f_\mathfrak{q}(v) \in K = \ker(f_\mathfrak{q}^m)$ then $f_\mathfrak{q}^{m+1}(v) = 0$, so $v \in \ker({f_\mathfrak{q}}^{m+1}) = K$.

- $K \cap x_1 M_\mathfrak{q} = x_1 K = f_\mathfrak{q}(K)$: "$\supseteq$" is trivial. For "$\subseteq$", if $w = f_\mathfrak{q}(v) \in K$ for some $v \in M_\mathfrak{q}$, then $v \in f_\mathfrak{q}^{-1} K = K$, so $w \in f_\mathfrak{q}(K) = x_1 K$.

For any endomorphism φ of a finite-length module N, additivity along $0 \to \ker\varphi \to N \to \varphi(N) \to 0$ and $0 \to \varphi(N) \to N \to N/\varphi(N) \to 0$ gives $\ell(\ker\varphi) = \ell(N/\varphi(N))$. We can apply this to $\varphi = f_\mathfrak{q}|_K$: since $\ker(f_\mathfrak{q}) \subseteq K$, we have $\ker(f_\mathfrak{q}|_K) = \ker(f_\mathfrak{q}) = V_\mathfrak{q}$, so with $f_\mathfrak{q}(K) = x_1 K$,

$$\ell(K/x_1K) = \ell(V_\mathfrak{q}).$$

On the other hand, the natural surjection $M_\mathfrak{q}/(x_1 M_\mathfrak{q}) \to (M_\mathfrak{q}/K)/(x_1(M_\mathfrak{q}/K))$ has the kernel $(K + (x_1 M_\mathfrak{q}))/(x_1 M_\mathfrak{q}) \cong K/(K \cap (x_1 M_\mathfrak{q})) = K/(x_1 K)$, giving (we now write $E = (M_\mathfrak{q}/K)$ for better readability) the exact sequence

$$0 \longrightarrow K/(x_1K) \longrightarrow U_\mathfrak{q} \longrightarrow E/(x_1E) \longrightarrow 0,$$

so by additivity and using $\ell(K/(x_1K)) = \ell(V_\mathfrak{q})$,

$$\ell(U_\mathfrak{q}) = \ell(V_\mathfrak{q}) + \ell\big(E/(x_1E)\big).$$

It remains to show the last term is positive.
Since $\mathfrak{q} \in \operatorname{Supp}_T U$, $U_\mathfrak{q} = M_\mathfrak{q}/(x_1 M_\mathfrak{q}) \neq 0$. We choose $\mathfrak{P} \in \operatorname{Supp}_{k[x,\xi]_\mathfrak{p}}(M_\mathfrak{q}/(x_1 M_\mathfrak{q}))$ and a minimal prime $\mathfrak{p} \subseteq \mathfrak{P}$ of $M_\mathfrak{q}$. We have shown above that $x_1 \notin \mathfrak{p}$, so x_1 is a unit in the localization at $\mathfrak{p}$; as K is annihilated by a power of x_1, $K_\mathfrak{p} = 0$. From $0 \to K_\mathfrak{p} \to (M_\mathfrak{q})_\mathfrak{p} \to E_\mathfrak{p} \to 0$ we get $E_\mathfrak{p} \cong (M_\mathfrak{q})_\mathfrak{p} \neq 0$, so $\mathfrak{p}$ lies in the support of $M_\mathfrak{q}/K$ in $k[x,\xi]_\mathfrak{p}$. Since support is closed under passing to larger primes [38, Lemma 10.40.5 (Tag 00L2)] and $\mathfrak{p} \subseteq \mathfrak{P}$, also $\mathfrak{P} \in \operatorname{Supp}_{k[x,\xi]_\mathfrak{p}} E$, i.e. $E_\mathfrak{P} \neq 0$.
Since $\mathfrak{P} \in \operatorname{Supp}_{k[x,\xi]_\mathfrak{p}}(M_\mathfrak{q}/(x_1 M_\mathfrak{q}))$ and $M_\mathfrak{q}/(x_1 M_\mathfrak{q})$ is annihilated by x_1, we have $x_1 \in \mathfrak{P}$. If $(x_1 E_\mathfrak{P}) = E_\mathfrak{P}$, then as x_1 lies in the maximal ideal of the local ring $(k[x,\xi]_\mathfrak{p})_\mathfrak{P}$, Nakayama's lemma [24], (see also [38, Tag 00DV]) would force $E_\mathfrak{P} = 0$, a contradiction. Hence $\big(E/(x_1E)\big)_\mathfrak{P} \neq 0$, so $E/(x_1E) \neq 0$ and $\ell\left(E/(x_1E)\right) > 0$.
Therefore $\ell(U_\mathfrak{q}) = \ell(V_\mathfrak{q}) + \ell\big(E/x_1E\big) > \ell(V_\mathfrak{q})$. □

Lemma 6.8. *The cokernel $U = M/x_1M$ of right multiplication by x_1 on $M = \operatorname{gr} Q$ vanishes, i.e. $U = 0$.*

Proof. If $U \neq 0$, it has a minimal support prime $\mathfrak{q}$. Lemma 6.5 makes the relevant lengths finite, and Lemmas 6.6 and 6.7 contradict one another. Thus $U = M/x_1M = 0$. □

7. Characteristics of the characteristic variety

Corollary 7.1. *Equip Q with the order filtration $Q_m := qF_m = \operatorname{im}(F_m \to Q)$ induced by the cyclic generator $q = [1]$, the class of 1 in Q. As before, we denote the symbol of $\tilde{d}$ in the order filtration by $P = \sigma_N(\tilde{d})$. Then*

$$P \in \operatorname{Ann}_{k[x,\xi]} \operatorname{gr} Q. \tag{7.1}$$

Proof. $Q_\bullet$ is a good filtration because Q is cyclic. Let $\bar{q}$ be the class of q in $\operatorname{gr} Q$. We know that $q\,\tilde{d} = [\tilde{d}] = 0$ in Q (because $\tilde{d} \in I$) and $\tilde{d}$ has order N (Lemma 4.5), so $q\tilde{d} \in Q_N$. Therefore $\bar{q} \cdot P$, its image in Q_N/Q_{N-1}, must also be zero, and hence P annihilates the class $\bar{q}$ of q in $\operatorname{gr} Q$. Since $\operatorname{gr} Q$ is generated over $k[x,\xi]$ by $\bar{q}$, we get $P \in \operatorname{Ann}_{k[x,\xi]} \operatorname{gr} Q$. □

Proposition 7.2. $\operatorname{Char}(Q) \cap \pi^{-1}(H) = \varnothing$.

Proof. Write $M = \operatorname{gr} Q$. Since M is finite over $k[x,\xi]$, $\operatorname{Supp}(M/x_1M) = \operatorname{Supp}(M) \cap V(x_1)$ [38, Tag 00L3], where $V(x_1)$ is the set of primes of $k[x, \xi]$ containing x_1. By Lemma 6.8, $M/x_1M = 0$, so its support is empty, hence $\operatorname{Supp}(M) \cap V(x_1) = \varnothing$. By definition $\operatorname{Supp}(M) = \operatorname{Char}(Q)$, and $V(x_1) = \pi^{-1}(H)$ is the preimage of $H = \{x_1 = 0\}$ under the projection π. Therefore $\operatorname{Char}(Q) \cap \pi^{-1}(H) = \varnothing$. □

Proof. We write $M = \operatorname{gr} Q$. Since M is finite over $k[x,\xi]$, its quotient by x_1 has support $\operatorname{Supp}_{k[x,\xi]}(M/x_1M) = \operatorname{Supp}_{k[x,\xi]}(M) \cap V(x_1)$ [38, Tag 00L3]. By Lemma 6.8 $M/x_1M = 0$ and has therefore no support, so $\operatorname{Supp}_{k[x,\xi]}(M) \cap V(x_1) = \varnothing$. Per definition $\operatorname{Supp}_{k[x,\xi]}(M) = \operatorname{Char}(Q)$, and $V(x_1) = \pi^{-1}(H)$ is the preimage of $H = \{x_1 = 0\}$ under the projection π. Therefore $\operatorname{Char}(Q) \cap \pi^{-1}(H) = \varnothing$.

□

Formal comparison: stated for the support transported along $\xi \mapsto -\xi$, which fixes x_1; transported back verbatim

8. A conormal direction at infinity

Throughout this section K denotes an algebraically closed field of characteristic zero. Let $Y \subset X = A^n_K$ be an irreducible closed subvariety and $\mathfrak{p} = I(Y)$ its prime ideal, so $Y = V(\mathfrak{p})$. Because $\mathbb{A}^n_K$ has a global coordinate frame, $T^*X = X \times (K^n)^*$ holds and we can therefore identify all cotangent fibres with $(K^n)^*$. We write $[Z_0 : \cdots : Z_n]$ for homogeneous coordinates on the projective space $\mathbb{P}^n$, and embed $X \hookrightarrow \mathbb{P}^n$ as the affine chart $\{Z_0 \neq 0\}$ via $[1 : x_1 : \cdots : x_n]$, so that $x_i = Z_i/Z_0$ and the hyperplane at infinity is $\{Z_0 = 0\}$. We define $\operatorname{Dir}(Y)$ to be the Zariski closure in $\mathbb{P}((K^n)^*)$ of the nonzero conormal vectors at all smooth affine points of Y projected onto $\mathbb{P}((K^n)^*)$.

Definition 8.1 (Asymptotic conormal directions)**.** *For Y as above, the set of* conormal directions *is*

$$\operatorname{Dir}(Y) = \overline{\{[\xi] \in \mathbb{P}((K^n)^*) : 0 \neq \xi \in N(Y)\}} \subseteq \mathbb{P}((K^n)^*),$$

the Zariski closure of the projectivization of $N(Y)$.

Let $\mathfrak{p} \subset K[x_1, \ldots, x_n]$ be prime and define $Y = V(\mathfrak{p})$; we assume $Y \cap H = \varnothing$. We furthermore write $R = K[x]/\mathfrak{p}$, $F = \operatorname{Frac}(R)$ for the field of fractions, and $\bar{x}_i$ for the image of x_i in R.

Lemma 8.2. *Let $\bar{x}_1$ be transcendental over K. Put $Z_0 = 1$, $Z_i = \bar{x}_i$ $(1 \le i \le n)$. There exist an index j, an element $s \in F^\times$, a finitely generated integrally closed K-subalgebra $A \subseteq F$ with $\operatorname{Frac} A = F$, and a height-one prime $\mathfrak{p}_D \subset A$ such that, with $q_i := Z_i s$ and $V := A_{\mathfrak{p}_D}$:*

(a) $q_j = 1$, $q_i \in A$ for all i, $\bar{x}_1 \in \mathfrak{p}_D$, and A is integral over $C_0 := K[q_0, \dots, q_n, \bar{x}_1]$;

(b) V is a DVR with $K \subseteq A \subseteq V$, $\operatorname{Frac} V = F$, $\bar{x}_1 \in \mathfrak{m}_V$, and $R \not\subseteq V$.

Proof. (1) A valuation ring. Since $\bar{x}_1$ is transcendental over K, $S = K[\bar{x}_1]$ is a polynomial ring and $\bar{x}_1 S$ is a proper ideal. By Chevalley's extension theorem there is a valuation ring $W \subseteq F$ with $S \subseteq W$ and $\bar{x}_1 \in \mathfrak{m}_W$. In particular $K \subseteq W$ and $\bar{x}_1^{-1} \notin W$.
(2) A chart. Among the nonzero Z_a choose j such that $Z_a/Z_j \in W$ for all a (possible since W is a valuation ring and the index set is finite), and put $s = Z_j^{-1}$. Then $q_j = 1$ and $q_a \in W$ for all a. Since $\bar{x}_i = q_i/q_0$ and $q_0 = s \neq 0$, the ring $C_0 = K[q_0, \dots, q_n, \bar{x}_1] \subseteq W$ is finitely generated with $\operatorname{Frac} C_0 = F$.
(3) Normalization. Let A be the integral closure of C_0 in F. It is finite over C_0 (finiteness of normalization for finitely generated domains over a field), hence a finitely generated, integrally closed, Noetherian domain with $\operatorname{Frac} A = F$. Since W is integrally closed, $A \subseteq W$, so $\bar{x}_1^{-1} \notin A$ and $\bar{x}_1$ is a nonzero nonunit of A.
(4) The divisor. By Krull's principal ideal theorem there is a prime $\mathfrak{p}_D \ni \bar{x}_1$ of A of height one. As A is normal and Noetherian, $V = A_{\mathfrak{p}_D}$ is a DVR with $\operatorname{Frac} V = F$, $A \subseteq V$ and $\bar{x}_1 \in \mathfrak{m}_V$. (Geometrically, V is the local ring of a prime divisor of the normalization of the affine chart $\{Z_j \neq 0\}$ of $\overline{Y}$. No properness is used.)
(5) If $R \subseteq V$, then $\bar{x}_1$ would be a unit of V because $\bar{x}_1^{-1} \in R$, contradicting $\bar{x}_1 \in \mathfrak{m}_V$. □

Lemma 8.3 (Residue field)**.** *Let $\kappa = V/\mathfrak{m}_V = \operatorname{Frac}(A/\mathfrak{p}_D)$ and $\bar{q}_i$ the residue of q_i. Then:*

(a) κ is algebraic over $L = K(\bar{q}_0, \dots, \bar{q}_n)$.

(b) $d := \dim_\kappa \Omega_{\kappa/K}$ is finite, and there are indices $i_1, \dots, i_d \notin \{0, 1, j\}$ such that $d\bar{q}_{i_1}, \dots, d\bar{q}_{i_d}$ form a basis of $\Omega_{\kappa/K}$. Let $D_1, \dots, D_d$ be the dual K-derivations of κ, so $D_k(\bar{q}_{i_l}) = \delta_{kl}$.

(c) $\Omega_{V/K}$ is a finitely generated V-module.

Proof. (a) Let $a' \in A$. As A is integral over $C_0 = K[q, \bar{x}_1]$, an integral equation of a' over C_0 reduces modulo $\mathfrak{m}_V$ to an integral equation of $\bar{a}'$ over the image of C_0 in κ. This image is $K[\bar{q}]$, because $\bar{x}_1 \in \mathfrak{m}_V$, so $\bar{a}'$ is algebraic over L. Every element of κ has the form $\bar{a}'/\bar{b}'$ with $a', b' \in A$, $b' \notin \mathfrak{p}_D$, and algebraic elements form a field.
(b) κ is a finitely generated field extension of K. Since $\bar{q}_0 = \bar{q}_1 = 0$ and $\bar{q}_j = 1$, the field L is generated by the remaining $\bar{q}_i$. Choose a transcendence basis $B = \{\bar{q}_{i_1}, \dots, \bar{q}_{i_d}\}$ of L/K among them; then $i_l \notin \{0, 1, j\}$. By (a), $\kappa/K(B)$ is algebraic, hence separable in characteristic 0, so $\Omega_{\kappa/K(B)} = 0$ and $d\bar{q}_{i_1}, \dots,$

$d\bar{q}_{i_d}$ is a basis of $\Omega_{\kappa/K}$. The dual basis of $\mathrm{Hom}_\kappa(\Omega_{\kappa/K},\kappa) = \mathrm{Der}_K(\kappa,\kappa)$ gives the D_k.
(c) A is a finitely generated K-algebra, so $\Omega_{A/K}$ is a finite A-module, and $\Omega_{V/K} = V \otimes_A \Omega_{A/K}$ because $V = A_{\mathfrak{p}_D}$ is a localization. □

Lemma 8.4 (Tangent space at y)**.** *Let $T_y = \bigcap_{f\in\mathfrak{p}} \ker df_y \subseteq \kappa((t))^n$ and let $\pi\colon \kappa((t))^{n+1} \to \kappa((t))^n$, $\pi(w)_i = (w_i q_0 - q_i w_0)/q_0^2$ $(1 \le i \le n)$, with kernel $\kappa((t))q$. Put $C = \pi^{-1}(T_y)$.*

- *(a) $\dim T_y = m \le d+1$; hence $\dim C = m+1 \le d+2$.*
- *(b) For every K-derivation δ of $\kappa((t))$ the column $\delta q = (\delta q_i)_i$ lies in C (in particular $q \in C$).*
- *(c) If a row vector $\ell \in \kappa((t))^{n+1}$ annihilates C, then $(\ell_1,\dots,\ell_n) \in \mathrm{span}\{df_y : f \in \mathfrak{p}\}$.*

Proof. (a) By the conormal sequence $\Omega_{F/K} = F^n/\mathrm{span}\{df(\bar{x})\}$, and $\dim_F \Omega_{F/K} = \mathrm{trdeg}_K F = m$ in characteristic 0 [18, II.8]. Rank of the Jacobian is preserved under $F \subset \kappa((t))$, so $\dim \mathrm{span}\{df_y\} = n-m$ and $\dim T_y = m$. For the bound: $\Omega_{V/K}$ is a finitely generated V-module ($V = A_{\mathfrak{p}_D}$), and the sequence $\mathfrak{m}_V/\mathfrak{m}_V^2 \to \kappa \otimes \Omega_{V/K} \to \Omega_{\kappa/K} \to 0$ gives $\dim_\kappa(\kappa \otimes \Omega_{V/K}) \le 1+d$. By Nakayama $\Omega_{V/K}$ is generated by $\le d+1$ elements, hence $m = \dim_F(F \otimes \Omega_{V/K}) \le d+1$. Since $\ker\pi$ is a line, $\dim C = \dim T_y + 1$.
(b) For $f \in \mathfrak{p}$, $f(y) = 0$ and δ kills K, so $0 = \delta f(y) = \sum_i \partial_i f(y)\,\delta y_i$, i.e. $\delta y \in T_y$. The quotient rule gives $\pi(\delta q)_i = \delta(q_i/q_0) = \delta y_i$.
(c) $\ell(q) = 0$ and $\ell(w) = 0$ for $w \in C$. For $w \in C$, $\sum_{i\ge1} \ell_i \pi(w)_i = \big[q_0 \sum_{i\ge1} \ell_i w_i - w_0 \sum_{i\ge1} \ell_i q_i\big]/q_0^2 = \big[-q_0\ell_0 w_0 + w_0 \ell_0 q_0\big]/q_0^2 = 0$. As $\pi(C) = T_y$, $(\ell_1,\dots,\ell_n) \in T_y^\perp = \mathrm{span}\{df_y\}$. □

Theorem 8.5. *Let $Y \subseteq \mathbb{A}^n$ be nonempty, closed, and irreducible with $Y \cap H = \varnothing$, and let $N(Y) = \bigcup_{y\in Y_{\mathrm{sm}}} N_y^* Y \subseteq K^n$ be the conormal cone over the smooth locus Y_{sm}.*

- *(i) Every polynomial $P \in K[\xi_1,\dots,\xi_n]$ vanishing on $N(Y)$ satisfies $P(1,0,\dots,0) = 0$.*
- *(ii) $[dx_1] \in \mathrm{Dir}(Y)$.*

Formal comparison: over every algebraically closed field of characteristic zero; the formal proof follows a different route, written out after the printed proof

Proof. If $\bar{x}_1$ is algebraic over K, there is a nonzero polynomial g over K in one variable with $g(\bar{x}_1) = 0$, i.e. $g(x_1) \in \mathfrak{p}$. Since K is algebraically closed, $g(x_1) = \lambda \prod_i (x_1 - c_i)$ with $\lambda \in K^\times$, and as $\mathfrak{p}$ is prime, some factor $x_1 - c$ lies in $\mathfrak{p}$. Thus $x_1 = c$ on Y, and $c \neq 0$ because $Y \neq \varnothing$ avoids $x_1 = 0$. In this case $f = x_1 - c \in \mathfrak{p} = I(Y)$ has differential $df_y = dx_1$ at every point $y \in Y_{\mathrm{sm}}$, so $dx_1 \in N_y^* Y$ and its coordinate vector in the dual basis $(1,0,\dots,0) \in N(Y)$. It follows directly that every polynomial vanishing on $N(Y)$ therefore vanishes at $(1,0,\dots,0)$, and $[dx_1] \in \mathrm{Dir}(Y)$. So we may assume

from now on that $\bar{x}_1$ is transcendental over K.

The rest of the proof follows this rough logic: We build an arc to a place at infinity where x_1 vanishes, with generic point y on Y. We show that the tangent planes $L(t)$ to Y at y, $t \neq 0$, have a limit plane L_0 of the same dimension at $t = 0$, and that $L_0 \subseteq \{Z_1 = 0\}$. Hence the covector $Z_1^\vee$ reading off the Z_1-coordinate annihilates L_0, and we can lift it to a family of covectors $\ell(t)$ annihilating $L(t)$ for $t \neq 0$, with $\ell(0) = Z_1^\vee$. For $t \neq 0$ the $\ell(t)$ are conormal covectors to Y, so their limit $\ell(0)$, whose affine part is dx_1, lies in the closure of the conormal directions, i.e. $[dx_1] \in \mathrm{Dir}(Y)$.

Since $\bar{x}_1$ is transcendental over K, by Lemma 8.2 there exists a divisorial valuation ring $V \subset F$ with $\bar{x}_1 \in \mathfrak{m}_V$ (the maximal ideal) and $R \not\subseteq V$. Let ϖ be a generator of $\mathfrak{m}_V$, then every nonzero $f \in F$ can be written as $f = \varpi^k \cdot u$ with $k \in \mathbb{Z}$ and u a unit of V - we say $\mathrm{ord}_V(f) = k \in \mathbb{Z}$ is the valuation of f in V. We denote the corresponding residue field by $\kappa = V/\mathfrak{m}_V$.
Let $\widehat{V}$ be the completion of V. Since $V \supseteq K$ and K is of characteristic zero, there exists a coefficient field of $\widehat{V}$ containing K and mapping isomorphically onto κ [38, Tag 0C0S]; choosing one gives use an isomorphism $\widehat{V} \cong \kappa[[t]]$ under which ϖ corresponds to the formal variable t. Since the natural map $V \to \widehat{V}$ is injective ($\bigcap_k \varpi^k V = 0$) and extends to an embedding $F \hookrightarrow \kappa((t))$, we identify elements of F with Laurent series.
Let $\bar{x}_0 := 1$. Choose $j \in \{0, \dots, n\}$ and a valuation ring W such that $\bar{x}_a/\bar{x}_j \in W$ for all a, and define $q_i := \bar{x}_i/\bar{x}_j$ for $0 \leq i \leq n$. Set $Z_0 = 1$, $Z_i = \bar{x}_i$ and let j and $q = (q_0, \dots, q_n)$ be as in Lemma 8.2, so that $q_i \in V$, $q_j = 1$, $q_1 = q_0 \bar{x}_1$ (since $q_1/q_0 = Z_1/Z_0 = \bar{x}_1$). Let $a = \mathrm{ord}_V(q_0)$ and $b = \mathrm{ord}_V(q_1)$. $a, b \geq 0$ since q_0, q_1 are elements of V, so that in $\kappa[[t]]$

$$q_0 = t^a u_0, \qquad q_1 = t^b u_1,$$

with $u_0(0) \neq 0 \neq u_1(0)$. Since $\bar{x}_1 \in \mathfrak{m}_V$, $b - a = \mathrm{ord}_V(q_1/q_0) = \mathrm{ord}_V(\bar{x}_1) > 0$. Moreover $a > 0$: if $a = 0$, then q_0 is a unit of V, hence all $\bar{x}_i = q_i/q_0$ lie in V, so $R \subseteq V$ since also $K \subseteq V$, contradicting Lemma 8.2(b). It follows that $j \notin \{0, 1\}$ since $q_j = 1$ is a unit while q_0, q_1 are not.
Dividing by $q_0 \neq 0$, we define $y = (q_1/q_0, \dots, q_n/q_0) \in \kappa((t))^n$; its coordinates are the images of $\bar{x}_1, \dots, \bar{x}_n$ under $F \hookrightarrow \kappa((t))$. Since ring homomorphisms commute with polynomial evaluation and this map is injective, for every polynomial $f \in K[x_1, \dots, x_n]$ we have

$$f(y) = 0 \iff f(\bar{x}_1, \dots, \bar{x}_n) = 0 \iff f \in \mathfrak{p}. \tag{8.1}$$

$f(y) = 0$ if and only if $f(\bar{x}_1, \dots, \bar{x}_n) = 0$ in F, i.e. if and only if $f \in \mathfrak{p}$.

We will now build independent vectors from the tangential planes of q. Let $d = \dim_\kappa \Omega_{\kappa/K}$, and let $i_1, \dots, i_d \notin \{0, 1, j\}$ and $D_1, \dots, D_d$ be as in Lemma 8.3.

We apply D_k coefficientwise to $q_i(t) = \sum_r c_{ir}t^r$ and define $Z_{ik}(t) := \sum_r D_k(c_{ir})t^r$. Since $D_k(\bar{q}_{i_l}) = \delta_{kl}$, the reduction $Z(0)$ is the identity on the rows $i_1, \dots, i_d$, and since $q_j = 1$ is constant, $Z_{jk} = 0$. Rows 0 and 1 of Z have order at least a and b, since D_k preserves t-adic order.
Let $\tilde{Z}$ be the $d \times d$ submatrix of Z consisting of the rows $i_1, \dots, i_d$ and all columns. Its reduction $\tilde{Z}(0)$ is the identity matrix, so $\tilde{Z}$ is invertible over $\kappa[[t]]$. Hence there exists a unique vector $\lambda = (\lambda_1, \dots, \lambda_d)^\mathsf{T} \in \kappa[[t]]^d$ with $\tilde{Z}\lambda = (q'_{i_1}, \dots, q'_{i_d})^\mathsf{T}$. We define $w = q' - Z\lambda$, so that $w_{i_l} = 0$ for $l = 1, \dots, d$. Its 0th entry is $w_0 = q'_0 - Z_{0\bullet}\lambda$. Here $q'_0 = at^{a-1}u_0 + t^a u'_0$ has order exactly $a-1$ (since the coefficient $au_0(0)$ of t^{a-1} is nonzero), while $Z_{0\bullet}\lambda$ has order at least a, because every entry of row 0 of Z has order at least a (since q_0 has order a). Hence $\operatorname{ord} w_0 = a-1$. Let c be the minimal order among the entries of w; then $c \le a-1$, and $\tau := t^{-c}w$ is a vector over $\kappa[[t]]$ with $\tau(0) \neq 0$ and $\tau_{i_l} = 0$. In row 1, both q'_1 and $Z_{1\bullet}\lambda$ have order at least $b-1$, so $\operatorname{ord} w_1 \ge b-1$, so $\operatorname{ord} \tau_1 \ge b-1-c \ge b-a > 0$. Finally $q'_j = 0$ and $Z_{j\bullet} = 0$ (as $q_j = 1$), so $\tau_j = 0$.

We can show that $q, Z_{\bullet 1}, \dots, Z_{\bullet d}, \tau$ span all of C: Let $M = [\, q \mid Z_{\bullet 1} \mid \cdots \mid Z_{\bullet d} \mid \tau\,]$, an $(n+1) \times (d+2)$ matrix over $\kappa[[t]]$. Its reduction $M(0)$ has full column rank: in row j only $q(0)$ is nonzero ($q_j = 1$); in rows $i_1, \dots, i_d$ the columns $Z_{\bullet k}(0)$ form an identity block while τ vanishes there; and $\tau(0) \neq 0$, i.e. we have $d+2$ independent reduced column vectors. This also translates to the column vectors of M themselves: Since some $(d+2) \times (d+2)$ minor of $M(0)$ is nonzero, the corresponding minor of M is a unit of $\kappa[[t]]$; hence the columns of M are also linearly independent over $\kappa((t))$, so they span a $(d+2)$-dimensional subspace there. Furthermore, every column of $M(0)$ has a vanishing Z_1-entry: $q_1(0) = 0$ (as $b > 0$), the entries Z_{1k} have order at least $b > 0$, and $\tau_1(0) = 0$. Thus the span L_0 of $q(0), Z_{\bullet 1}(0), \dots, Z_{\bullet d}(0), \tau(0)$ is contained in $\{Z_1 = 0\}$ and also has dimension $d+2$.
By Lemma 8.4(b), the columns q, $Z_{\bullet k}$ and q' lie in C (each is δq for a K-derivation δ of $\kappa((t))$: the D_k extended coefficientwise, and d/dt), hence so does τ, a combination of them. By Lemma 8.4(a), $\dim C \le d+2$, so the $d+2$ independent columns of M must span C.

Let $u = (0, 1, 0, \dots, 0)$ be the row vector reading off the Z_1-coordinate; since $L_0 = \operatorname{im} M(0) \subseteq \{Z_1 = 0\}$ by Step 3, $uM(0) = 0$. We can lift u to a row vector $\ell(t) \in \kappa[[t]]^{n+1}$ with $\ell M = 0$ and $\ell(0) = u$: Since $M(0)$ has full column rank $d+2$, there is a set I of $d+2$ row indices for which the square submatrix $M_I(0)$ of $M(0)$ is invertible; then M_I is invertible over $\kappa[[t]]$, since its determinant has a nonzero constant term. Let J be the complementary set of row indices, and define ℓ by

$$\ell_J(t) = u_J, \qquad \ell_I(t) = -u_J M_J M_I^{-1},$$

where M_J denotes the submatrix of M with rows in J. Then $\ell(t)M = \ell_I(t)M_I + \ell_J(t)M_J = -u_J M_J + u_J M_J = 0$. Moreover $uM(0) = 0$ gives $u_I = -u_J M_J(0) M_I(0)^{-1}$, hence $\ell(0) = u$.

Since the columns of M span C, ℓ annihilates C, so by Lemma 8.4(c) its affine part $\tilde{\ell}(t) = (\ell_1(t), \dots, \ell_n(t))$ lies in $\operatorname{span}\{df_y : f \in \mathfrak{p}\}$, and since $\ell(0) = u$, $\tilde{\ell}(0) = (1, 0, \dots, 0)$.
Finally, let $P \in K[\xi_1, \dots, \xi_n]$ vanish on $N(Y)$. We will show $P(\tilde{\ell}(0)) = 0$: Since $\tilde{\ell} \in \operatorname{span}_{\kappa((t))}\{dg_y : g \in \mathfrak{p}\}$, we can write

$$\tilde{\ell} = \sum_{k=1}^{s} \mu_k \nabla g_k(y), \qquad g_k \in \mathfrak{p},\ \mu_k \in \kappa((t)),$$

With some $s \in \mathbb{N}$. Here $\nabla g(x) = (\partial g/\partial x_1(x), \dots, \partial g/\partial x_n(x))$ denotes the coordinate vector of the covector dg_x. We furthermore define

$$\Phi(x, \mu) = P\Big(\sum_{k=1}^{s} \mu_k \nabla g_k(x)\Big) = \sum_{\alpha} \phi_\alpha(x)\mu^\alpha \in K[x_1, \dots, x_n, \mu_1, \dots, \mu_s],$$

with $\phi_\alpha \in K[x_1, \dots, x_n]$ and $\mu^\alpha = \mu_1^{\alpha_1} \dots \mu_s^{\alpha_s}$ for a multi-infex $\alpha \in \mathbb{N}^s$. For a smooth point $y' \in Y_{\mathrm{sm}}$ and $\mu \in K^s$ we have $\sum_k \mu_k dg_k(y') \in N^*_{y'}Y \subseteq N(Y)$, so $\Phi(y', \mu) = 0$. As K is infinite, a polynomial in μ vanishing on all of K^s is zero, so $\phi_\alpha(y') = 0$ for all α. Since Y_{sm} is dense in Y, every ϕ_α vanishes on all of Y, i.e. $\phi_\alpha \in \mathfrak{p} = I(Y)$. By (8.1) (applied to the generic point $y \in \kappa((t))^n$), we therefore have $\phi_\alpha(y) = 0$ for all α. Hence $\Phi(y, \mu) = \sum_\alpha \phi_\alpha(y)\mu^\alpha = \sum_\alpha 0 \cdot \mu^\alpha = 0$ for all $\mu \in \kappa((t))^s$, and in particular

$$P(\tilde{\ell}) = \Phi\big(y, (\mu_1, \dots, \mu_s)\big) = 0.$$

Now $\tilde{\ell} \in \kappa[[t]]^n$ and P has coefficients in K, so $P(\tilde{\ell}) \in \kappa[[t]]$, and its constant term is $P(\tilde{\ell}(0))$. Since $P(\tilde{\ell}) = 0$, we get $P(\tilde{\ell}(0)) = P(1, 0, \dots, 0) = 0$, which is statement (i).
We have shown that every polynomial P vanishing on $N(Y)$ vanishes at $(1, 0, \dots, 0)$ (statement (i)). By the definition of the Zariski closure, $\mathrm{Dir}(Y)$ is the common zero set of all homogeneous polynomials vanishing on $\{[\xi] : 0 \neq \xi \in N(Y)\}$, equivalently on $N(Y)$. Each such polynomial vanishes at $(1, 0, \dots, 0)$, hence at its class $[dx_1] = [1 : 0 : \dots : 0]$ (homogeneity makes the vanishing independent of the representative). Therefore $[dx_1] \in \mathrm{Dir}(Y)$. □

Formal comparison: any algebraically closed field of characteristic 0; (i) is the affine-cone form used in Section 9

9. The involutive obstruction

Definition 9.1 (Fibre-conical)**.** *A closed subset $W \subseteq T^*X$ is* fibre-conical *if it is stable under the fibrewise scaling $(y, \xi) \mapsto (y, t\xi)$, $t \in \mathbb{G}_m$.*

Theorem 9.2 (Hilbert's Nullstellensatz)**.** *Let K be an algebraically closed field and $J \subseteq K[z_1, \dots, z_N]$ an ideal. Then*

(a) $V(J) = \varnothing \iff 1 \in J$
(b) $I(V(J)) = \sqrt{J}$

Theorem 9.3 (Involutive exclusion)**.** *Let K be an algebraically closed field of characteristic zero, and let $W \subseteq T^*X$ be a nonempty Zariski closed subset that is fibre-conical and involutive. Let $P \in K[\xi_1, \dots, \xi_n]$ be a polynomial with $P(1, 0, \dots, 0) \neq 0$. If P vanishes on W, then $W \cap \pi^{-1}(H) \neq \varnothing$.*

Formal comparison: any algebraically closed field of characteristic 0; P need not be homogeneous

Proof. Suppose we have a nonempty closed subset $W \subseteq T^*X$ that is fibre-conical and involutive. Suppose further, that $W \cap \pi^{-1}(H) = \varnothing$. We define the (radical) vanishing ideal $J = I(W) \subseteq K[x, \xi]$ consisting of all polynomials that vanish at every point $(y, \xi) \in W$, and $B = J \cap K[x]$, consisting of the functions $f(x)$ with $f(y) = 0$ for every $(y, \xi) \in W$.

Because W is fibre-conical, the ξ-homogeneous components of any polynomial in J again lie in J: if $g = \sum_d g_d \in J$ and $(y, \xi) \in W$, then $\sum_d a^d g_d(y, \xi) = g(y, a\xi) = 0$ for all $a \in K^\times$, and a polynomial in a with infinitely many roots is zero, so every $g_d(y, \xi) = 0$. Hence J is ξ-homogeneous.

Let $g \in J$ be written in its ξ-homogeneous components form, that all lie in J. In particular, also $g_0 \in J$. Since g_0 has ξ-degree 0, it is a polynomial in x alone, so $g_0 \in B$. Let $Y = V(B)$. For $y \in Y$ we therefore have $g(y, 0) = g_0(y) = 0$, so every $g \in J$ vanishes at $(y, 0)$, i.e. $(y, 0) \in V(J)$. Since W is closed, $V(J) = V(I(W)) = W$, hence $(y, 0) \in W$, or $Y \times \{0\} \subseteq W$. Furthermore, Y is the projection of W and is therefore per definition of Y closed: $\pi(W) \subseteq Y$ since for $(y', \xi') \in W$ and $f \in B \subseteq J$, we have $f(y') = f(y', \xi') = 0$ so $y' \in V(B) = Y$, and $Y \subseteq \pi(W)$, since if $y \in Y$, then also $(y, 0) \in W$, so $y = \pi(y, 0) \in \pi(W)$.
Lastly, assume there exists a $y \in Y \cap H$, then $(y, 0) \in W$ and $\pi(y, 0) = y \in H$, which would put $(y, 0) \in W \cap \pi^{-1}(H)$, which is wrong by our assumption. Therefore, $Y \cap H = \varnothing$.

Let $f \in B$. Then $f \circ \pi \in J$. The Hamiltonian derivation of f applied to a $a \in J$ is

$$H_f g = \sum_i \frac{\partial f}{\partial x_i} \frac{\partial g}{\partial \xi_i} = \{f, g\}$$

since the second term of the Poisson bracket vanishes because f does not depend on ξ. Since $f, g \in J$ and $\{J, J\} \subseteq J$ by the involutivity of W, it follows that $H_f g \in J$ and by induction also $H_f^m g \in J$. For a fixed $y \in Y$ and $g \in J$, we define the polynomial $\varphi(t) = g(y, t\, df_y) \in K[t]$. Its derivative is $\varphi'(t) = \sum_i \partial_{x_i} f(y) \frac{\partial g}{\partial \xi_i}(y, t df_y) = H_f g(y, t df_y)$, so by iteration, $\varphi^{(m)}(t) = H_f^m g(y, t df_y)$ for $m \geq 0$. Evaluating at $t = 0$, $\varphi^{(m)}(0) = H_f^m g(y, 0)$. Since we have established that $(y, 0) \in W$ and $H_f^m g \in J$, and all polynomials in J vanish on W, we have $\varphi^{(m)}(0) = 0$ for all $m \geq 0$ – so $\varphi(t)$ is a polynomial

it t all of whose derivatives vanish at $t = 0$. If we write $\varphi(t) = \sum_m c_m t^m$, then $\varphi^{(m)}(0) = m!c_m$ and since $m! \neq 0$ because K has characteristic zero, $c_m = 0$ for all m, and therefore, $\varphi(t) = 0$ everywhere. Remembering the definition of $\phi = g(y, t\,df_y) = 0$, so $(y, t\,df_y) \in V(J) = W$. More generally, since any linear combination of elements in B is again in B, we can apply this result to $f = \sum_a c_a f_a \in B$ and evaluating at $t = 1$, it follows that

$$(y, df_y) = (y, d(\sum_a c_a f_a)_y) = (y, \sum_a c_a d(f_a)_y) \in W,$$

so the whole linear span $\{y\} \times \operatorname{span}\{df_y : f \in B\}$ is a subset of W.

Since $V(B) = Y = \pi(W) \neq \varnothing$, $B \neq K[x]$, so B must have at least one minimal prime. The ideal B is radical (by construction since it is a vanishing ideal) in the Noetherian ring $K[x]$, so B has only finitely many minimal primes $\mathfrak{q}_1, \ldots, \mathfrak{q}_r$ and $B = \bigcap_i \mathfrak{q}_i$. We choose $\mathfrak{q} = \mathfrak{q}_1$ and set $Y_0 = V(\mathfrak{q})$. Since $\mathfrak{q}$ is prime, $1 \notin \mathfrak{q}$, so $Y_0 \neq \varnothing$ by the Nullstellensatz (Theorem 9.2); moreover Y_0 is closed and irreducible since $\mathfrak{q}$ is prime [18, Cor. I.1.4], and $Y_0 \subseteq Y$, hence $Y_0 \cap H = \varnothing$, as we have already established $Y \cap H = \varnothing$.
Furthermore, if $r = 1$, we choose $s = 1$, then $sh \in \mathfrak{q} = B$, and if $r \geq 2$, choose for every $i \geq 2$ elemens $s_i \in \mathfrak{q}_i \setminus \mathfrak{q}$. Then $s = s_2 s_3 \ldots s_r$ does not lie in $\mathfrak{q}$ since no factor lies in $\mathfrak{q}$, but does lie in every other $\mathfrak{q}_i$ since they are ideals and one of the factors lies in them. On the other hand, if $h \in \mathfrak{q}$, then $sh \in \mathfrak{q}$ since $h \in \mathfrak{q}$, and $sh \in \mathfrak{q}_i$ since $s \in \mathfrak{q}_i$ for all $i \geq 2$. Therefore, $sh \in \bigcap_i \mathfrak{q}_i = B$, i.e. $s\mathfrak{q} \subseteq B$. We now fix such a $h \in \mathfrak{q}$, and define $Y_0 = V(\mathfrak{q})$. For a $y \in Y_0 \cap \{s \neq 0\}$, $(y, dh_y) \in W$: Since $sh \in B$, and $d(sh)_y = s(y)dh_y + h(y)ds_y = s(y)dh_y$ since $h(y) = 0$, so $(y, s(y)dh_y) \in W$ and since $s(y) \neq 0$, also $(y, dh_y) \in W$. For $g \in J$, the polynomial $G(x) = g(x, dh_x)$ therefore vanishes on $Y_0 \cap \{s \neq 0\}$, and sG vanishes on all of Y_0, so $sG \in \mathfrak{q}$ and therefore $G \in \mathfrak{q}$, which means that G vanishes on all of Y_0. Since this is true for every $g \in J$, we have $g(y, dh_y = G(y) = 0$ for all $g \in J$, so $(y, dh_y) \in V(J) = W$. As before, we can argue that since $\mathfrak{q}$ is an ideal, it is closed under K-linear combinations, so for $h_1, \ldots, h_k \in \mathfrak{q}$ and $c_a \in K$, also $(y, \sum_a c_a (dh_a)_y) \in W$, or put differently, $\{y\} \times \operatorname{span}\{dh_y : h \in \mathfrak{q}\} \subseteq W$ for all $y \in Y_0$. At a smooth point y of Y_0, by Definition **??**, the conormal space is

$$N_y^* Y_0 = \operatorname{span}\{df_y : f \in \mathfrak{q}\}$$

since $\mathfrak{q}$ is prime and hence radical, and so $\mathfrak{q} = I(Y_0)$. We therefore have that $N_y^* Y_0 \subseteq \{y\} \times W$ for every smooth $y \in Y_0$, and hence $N(Y_0) \subseteq W$.
By assumption, P vanishes on W and hence on $N(Y_0)$. Because we have established, that Y_0 is closed, irreducible, and $Y_0 \cap H = \varnothing$, Y_0 therefore fulfills all requirements of Theorem 8.5, so $P(1, 0, \ldots, 0) = 0$, contradicting our initial assumption. Therefore, $W \cap \pi^{-1}(H) \neq \varnothing$. □

10. Proof of Theorem 1.1

Lemma 10.1. *Let K/k be a field extension and put $I_K = \tilde{d}A_n(K) + x_1^N \tilde{d}A_n(K)$, $Q_K = Q \otimes_k K = A_n(K)/I_K$, with the order filtration $Q_{K,i} = q\,F_iA_n(K)$. Then*

$$\operatorname{gr} Q_K \cong \operatorname{gr} Q \otimes_k K, \qquad \operatorname{Ann}_{K[x,\xi]} \operatorname{gr} Q_K = \big(\operatorname{Ann}_{k[x,\xi]} \operatorname{gr} Q\big)\, K[x,\xi],$$

and $Q = 0 \iff Q_K = 0$. In particular, $\operatorname{Char}(Q_K)$ is the preimage of $\operatorname{Char}(Q)$ under $\operatorname{Spec} K[x,\xi] \to \operatorname{Spec} k[x,\xi]$.

Proof. By the PBW basis, $F_iA_n(K) = F_iA_n(k) \otimes_k K$ and $I_K = I \otimes_k K$, so $Q_{K,i} = Q_i \otimes_k K$. As K is flat over k, $Q_{K,i}/Q_{K,i-1} = (Q_i/Q_{i-1}) \otimes_k K$, and summing over i gives $\operatorname{gr} Q_K = \operatorname{gr} Q \otimes_k K$. The module $\operatorname{gr} Q$ is finitely generated (cyclic, generated by the class of q) over the Noetherian ring $k[x,\xi]$, and $k[x,\xi] \to K[x,\xi]$ is flat, so the annihilator of the finite module commutes with this base change [38, Tag 05DT]. Finally, $Q \otimes_k K = 0$ implies $Q = 0$, because K is a nonzero free k-module. □

Proof of Theorem 1.1. Let $d \in A_n(k)$ be nonzero of Bernstein degree N and let $\tilde{d} = c^{-1}\Phi_g(d)$ be its monic normal form (Remark 4.3). If $n = 0$, then $A_0(k) = k$ and every nonzero d is a unit anyways, so let $n \geq 1$. As in Section 5 we define

$$I = \tilde{d}A_n(k) + x_1^N \tilde{d}A_n(k), \qquad Q = A_n(k)/I.$$

It suffices to show $Q = 0$: then $1 \in I$, i.e. there are $R, S \in A_n(k)$ with $1 = \tilde{d}R + x_1^N \tilde{d}S$.
Let $K = \overline{k}$, and let

$$W = \operatorname{Char}(Q_K) = V\big(\operatorname{Ann}_{K[x,\xi]} \operatorname{gr} Q_K\big) \subseteq K^n \times K^n,$$

with vanishing ideal $I(W) = \sqrt{\operatorname{Ann} \operatorname{gr} Q_K}$ by the Nullstellensatz (Theorem 9.2). We check the hypotheses of Theorem 9.3 for W and the polynomial P of Section 7:

- W is Zariski closed, since it is the zero set of an ideal of polynomials.
- The module $\operatorname{gr} Q_K$ is graded by the order filtration, which assigns degree 0 to the x_i and degree 1 to the ξ_i; so $\operatorname{Ann} \operatorname{gr} Q_K$ is homogeneous in ξ. Hence if $(y,\xi) \in W$ and $g \in \operatorname{Ann} \operatorname{gr} Q_K$, then $g(y, t\xi) = \sum_d t^d g_d(y,\xi) = 0$ for every $t \in K^\times$, where g_d denotes the ξ-homogeneous component of degree d of g. So $(y, t\xi) \in W$, i.e. W is fibre-conical.
- By Gabber's theorem (Theorem 3.7) $I(W)$ is involutive.
- By eq. (7.1), $P \in \operatorname{Ann}_{k[x,\xi]} \operatorname{gr} Q$, hence $P \in \operatorname{Ann}_{K[x,\xi]} \operatorname{gr} Q_K$ by Lemma 10.1, so P vanishes on W.
- By eq. (4.3), $P(1,0,\dots,0) = 1 \neq 0$.
- $W = \operatorname{Char}(Q_K)$ is the preimage of $\operatorname{Char}(Q)$ under $\operatorname{Spec} K[x,\xi] \to \operatorname{Spec} k[x,\xi]$ (Lemma 10.1), and $\pi^{-1}(H) = V(x_1)$ over K is the preimage of $\pi^{-1}(H) = V(x_1)$ over k. By Proposition 7.2, $\operatorname{Char}(Q) \cap$

$\pi^{-1}(H) = \varnothing$, and the preimage of an empty set is also empty, therefore $W \cap \pi^{-1}(H) = \varnothing$.

By Theorem 9.3, $W = \varnothing$. As K is algebraically closed, by the Nullstellensatz (Theorem 9.2) $1 \in \mathrm{Ann}_{K[x,\xi]} \operatorname{gr} Q_K$, so $\operatorname{gr} Q_K = 0$. Since the filtration of Q_K is exhaustive with $Q_{K,-1} = 0$, induction on m implies $Q_{K,m} = 0$ for all m, so $Q_K = 0$, and $Q = 0$ by Lemma 10.1. Therefore, $A_n(k) = \tilde{d}A_n(k) + x_1^N \tilde{d}A_n(k)$, or equivalently $1 = \tilde{d}R + x_1^N \tilde{d}S$ with $R, S \in A_n(k)$.

Lastly, we apply the inverse of the automorphism Φ_g of Remark 4.3. Since $\tilde{d} = c^{-1}\Phi_g(d)$ with $c \in k^\times$ central, this gives

$$1 = d\big(c^{-1}\Phi_g^{-1}(R)\big) + F\, d\big(c^{-1}\Phi_g^{-1}(S)\big), \qquad F = \Phi_g^{-1}(x_1)^N,$$

which proves Theorem 1.1. □

Formal comparison: stated for the canonical ideal; the argument uses nothing else

Lastly, as Stafford already showed, from $1 = dR + FdS$ follows cyclicity of the torsion right module:

Corollary 10.2. *Every finitely generated torsion right $A_n(k)$-module is cyclic.*

Proof. Suppose for every $d \neq 0 \in A_n$, there exists $F, R, S \in A_n$, such that $1 = dR + FdS$. Let us denote a torsion module by T. Stafford's theorem (Theorem 3.7 [31, p. 438]) established, that every finitely generated torsion module can be generated by 2 generators, so let us denote the generators of T with t_1 and t_2. We can define a homomorphism

$$\phi : A_n \oplus A_n \to T \quad \phi(r, s) \mapsto t_1 r + t_2 s.$$

This map is surjective, since t_1, t_2 generate all of T. By the first isomorphism theorem (for $\phi : M \to N$, $im(\phi) \cong M/ker(\phi)$), it follows

$$T \cong (A_n \oplus A_n)/M$$

with $M = ker(\phi)$.

T is torsion and therefore $\forall t \in T : \exists m \in A_n \backslash \{0\} : tm = 0$. Specifically, let us denote with d_1 and d_2 the annihilators of t_1 and t_2, i.e.

$$t_1 d_1 = t_2 d_2 = 0.$$

A_n is a right Ore domain, which means it has the property that for any nonzero d_1, d_2 there exists elements $r_1, r_2 \in A_n$ such that $d_1 r_1 = d_2 r_2 \neq 0$. Defining therefore $d := d_1 r_1 = d_2 r_2$ yields us a common annihilator of t_1 and t_2.

$(d, 0)$ and $(0, d)$ are elements in $M = ker(\phi)$, since $\phi(d, 0) = t_1 d = 0$ and $\phi(0, d) = t_2 d = 0$.

Because $(0, d), (d, 0) \in M$, also $(d, 0)S + (0, d)R = (dS, dR) = (dS, 1 - FdS) =: (a, 1 - Fa) \in M$, i.e. $(a, 1 - Fa)$ is an element of M.

On the other hand, $A_n \oplus A_n$ can be generated by $(a, 1 - Fa)$ and $(1, -F)$. We can check this by finding u, v such that for all $(x, y) \in A_n \oplus A_n$ we can write

$$(x, y) = (a, 1 - Fa)u + (1, -F)v.$$

Suitable choices are $u = y + Fx$ and $v = x - a(y + Fx)$.
Lastly, since $A_n \oplus A_n$ can be generated by $(a, 1 - Fa)$ and $(1, -F)$ and $(a, 1 - Fa)$ is an element of M, it follows that $T \cong (A_n \oplus A_n)/M$ can be generated by the image of $(1, -F)$, i.e. T is cyclic. □

11. Outlook

~~We have proved~~ The cofactors R, S are obtained non-constructively: the argument proves $Q = 0$ without producing a certificate. Effective bounds for $\deg_B R$ and $\deg_B S$ in terms of $\deg_B d$, and an algorithm in the spirit of the constructive two-generator methods, remain open. The Global Stafford extension raises the same questions for rings of differential operators on smooth affine varieties.

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Appendix A. AI-assisted research and formalization

Proof discovery, counterexample searches, review, and formalization were assisted by agentic AI tools. The work began with cyclic-extension computations in Macaulay2 and Maple. Attempts to construct universal cofactors directly produced special cases and reductions but eventually stalled. The decisive change was to study the canonical quotient $Q = A_n/(dA_n + x^N dA_n)$: Euler surjectivity and monicity constrained its characteristic support, while Gabber's involutivity and asymptotic conormal geometry forced that support to be empty. This turned cofactor selection into a quotient-vanishing problem. Full Lean formalization was deferred until the paper argument was complete, although small formal checks continued throughout. In the final formal

development, support avoidance over the coordinate hyperplane was proved by a localized filtered Koszul length contradiction.

The workflow separated proposed ideas from verified claims. One controller maintained the authoritative proof state, while parallel workers explored distinct mechanisms and returned evidence with an explicit scope. Independent reviewers checked fixed revisions for quantifiers, sidedness, circularity, and missing hypotheses. The project protocols were based on a `math-frontier` agent skill developed for this purpose: freeze the claim, identify its first unresolved dependency, and specify how a route could fail. Conceptual prompts requested a small number of distinct mechanisms; paper-only tasks sought complete arguments before implementation. A condensed representative prompt, adapted from the archived protocols, is:

> Work on one precisely stated unresolved claim. Propose at most three distinct mechanisms, each with an intermediate lemma, a noncircular justification, and an adversarial test. Preserve the quantifiers and right-sided cofactors. For a paper task, supply a complete argument; for formalization, prove the required hypotheses without adding axioms or using `sorry`. Return evidence and the first unsupported bridge; do not promote a conditional result to a theorem.

Research and review sessions used GPT-5.5, GPT-5.6 Luna and Sol, GPT-6 Astra, and Claude Fable 5, Haiku 4.5 and Opus 5. The project protocols assigned routine implementation and replay to Luna and conceptual consultation to Sol; independent reports also record Opus reviews. Subsequent manuscript, proof-map, build and release work used Fable 5.1, Opus 5, Sonnet 5 and GPT-6 Luna. These records establish participation and task roles, without assigning a particular mathematical idea to one model.

The completed proof uses Lean 4 and Mathlib and pins AlgebraicAnalysis v0.3.0 [16, 37, 3]; the later library archive is v0.3.3 [4]. Formalization distinguished theorems proving required inputs from conditional theorems merely consuming them. Statement comparison and independent NanoDa and Lean kernel checks verified the submitted declarations [21, 25, 26]; their proofs use only `propext`, `Classical.choice`, and `Quot.sound`. The Stafford38 and Global Stafford headline proofs are registered separately in Palomar [7, 5]; their Lean developments are archived [8, 6].

Appendix B. Further work

AI comment REF-01. Proposed change — author acceptance pending. Please confirm that the references here point to the order-symbol result and the characteristic-containment statement used in the argument.

~~AI also proved some related stuff regarding Right Ore localizations, the corresponding left-hand identity, and apparently the corresponding identity over other rings such as differential rings with laurent series coefficients; I am leaving that out for now.~~ The formalization also proves the following consequences, which are not otherwise used here: the identity in every right Ore

localization of $A_n(k)$; the left-handed identity $1 = Rd + SdF$ via the formal adjoint ; the identity for finite-order differential operators on localizations of $k[x_1, \ldots, x_n]$, including principal opens, Laurent-polynomial coefficients and the fraction field ; and, independently of the geometric argument, an evolutionary version with arbitrary coefficients commuting with one Weyl pair .

AI comment SCOPE-01. Proposed change — author acceptance pending. Review the scope before accepting this paragraph. The coefficients in the stated localization result are Laurent polynomials, not arbitrary Laurent series, and the operators have finite order. Retain further consequences only with their exact hypotheses and proof references; optional formal declaration links are available in the technical supplement.

Graz University of Technology